



SOICT

HUST
ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

Multimodal Machine Learning

Lecture 5.1: Multimodal Transformers (Part 1)

Dam Quang Tuan

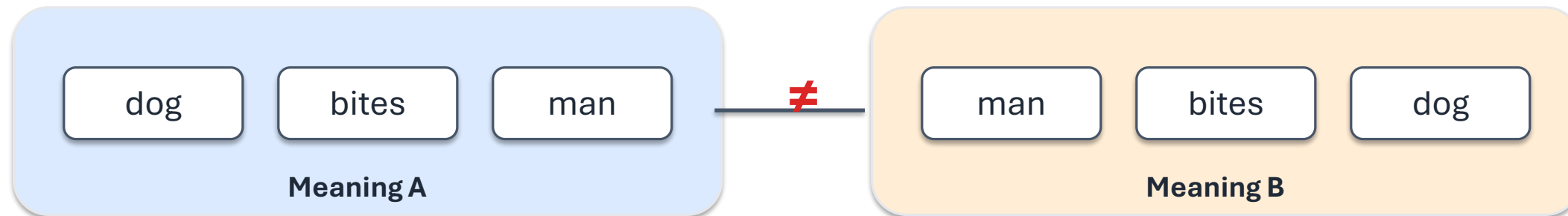
Objectives of today's class

- Positional embeddings
- Language pre-training
 - BERT: Bidirectional Encoder Representations from Transformers
- Multimodal transformers (Image and language)
 - Concatenated transformers (VisualBERT, Uniter)
 - Crossmodal transformers (ViLBERT, LXMERT)
 - Modality-shift transformer (MAG-BERT)
- Sequence-to-sequence modeling with Transformers

Positional Embeddings

Why Word Order Matters

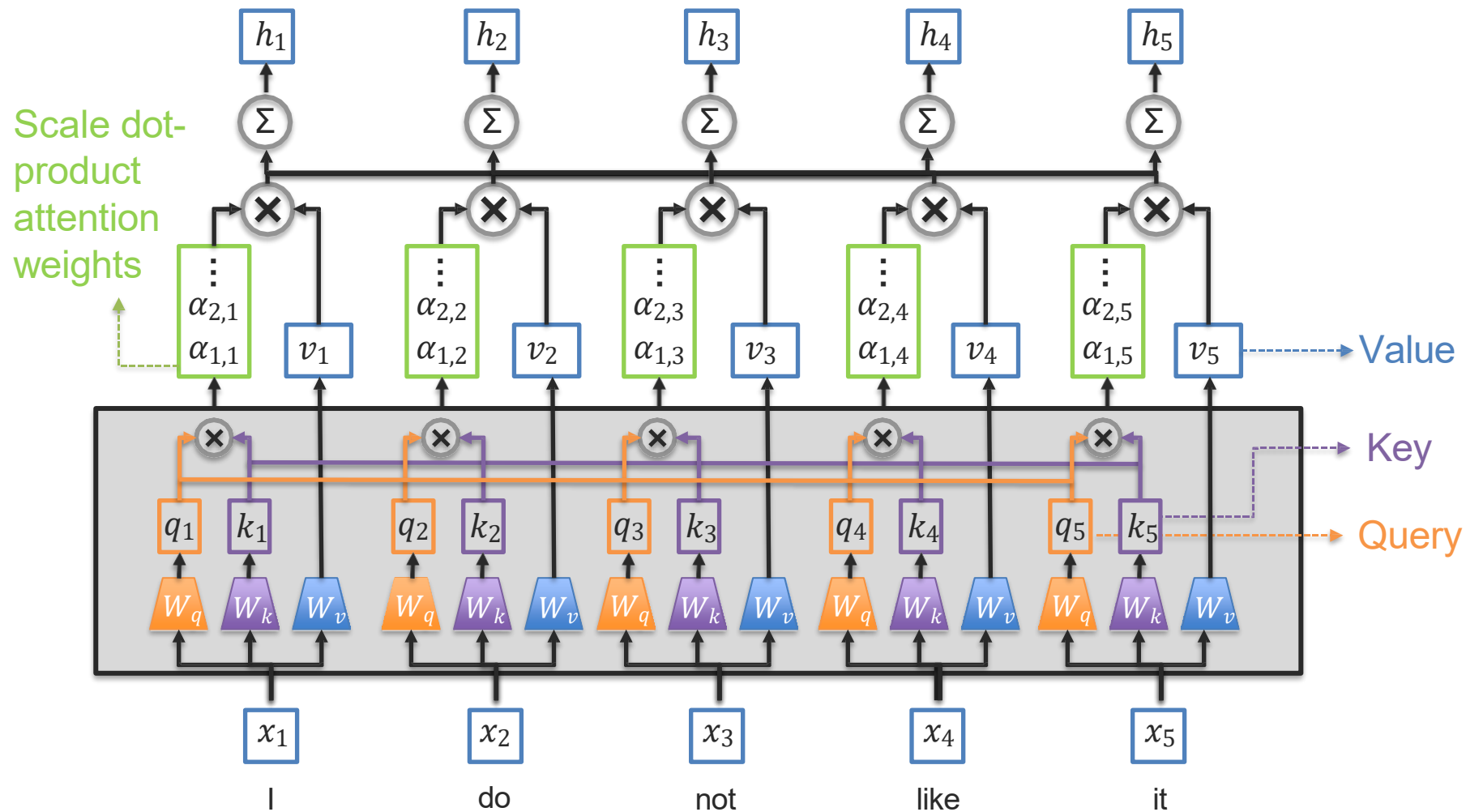
Same words. Different order. Different meaning.



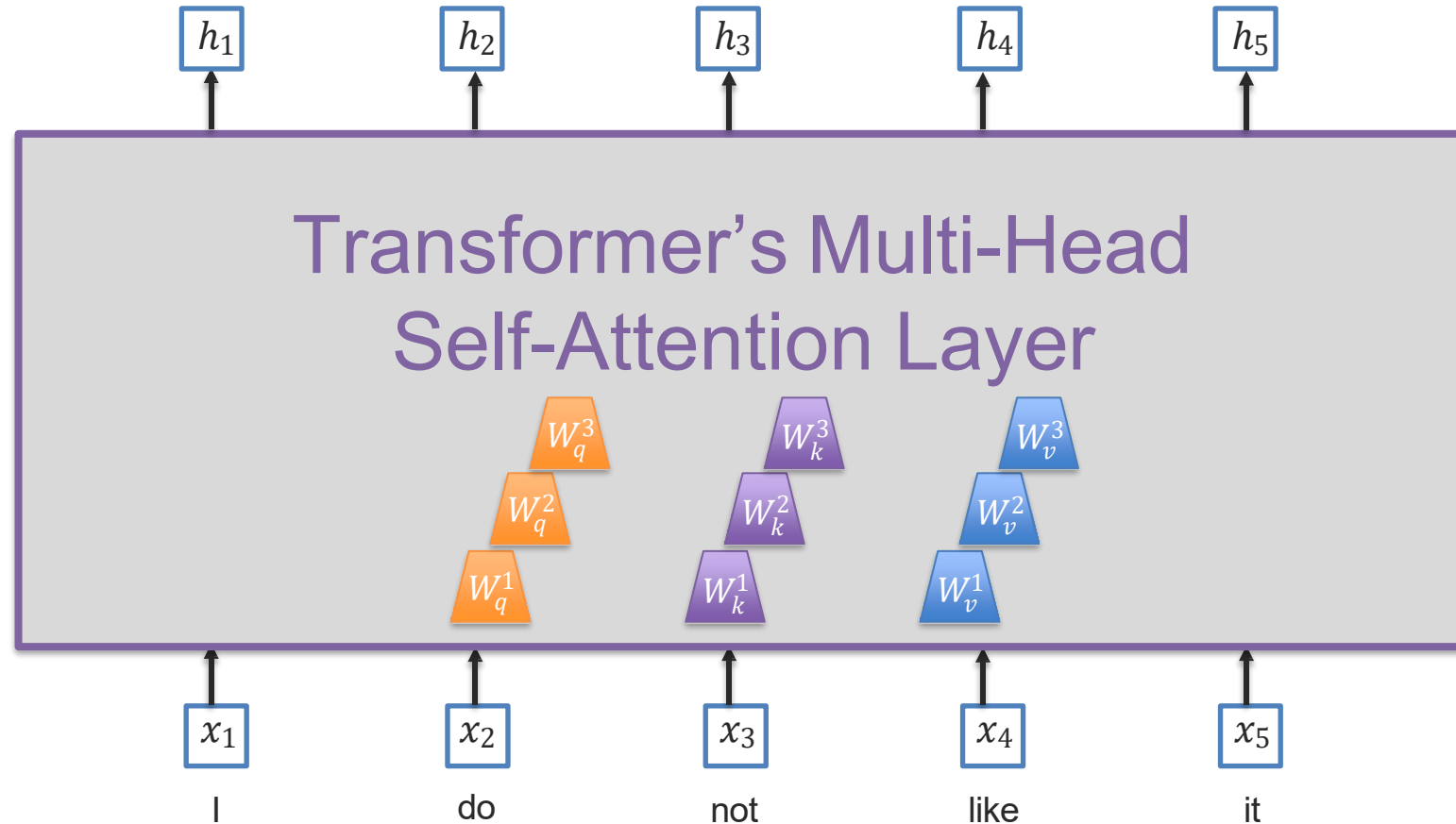
Self-attention without position information treats reordered inputs as the same pattern, only permuted.

Self-attention mixes information across tokens
But token order is not encoded by default
Position embeddings inject order into the input

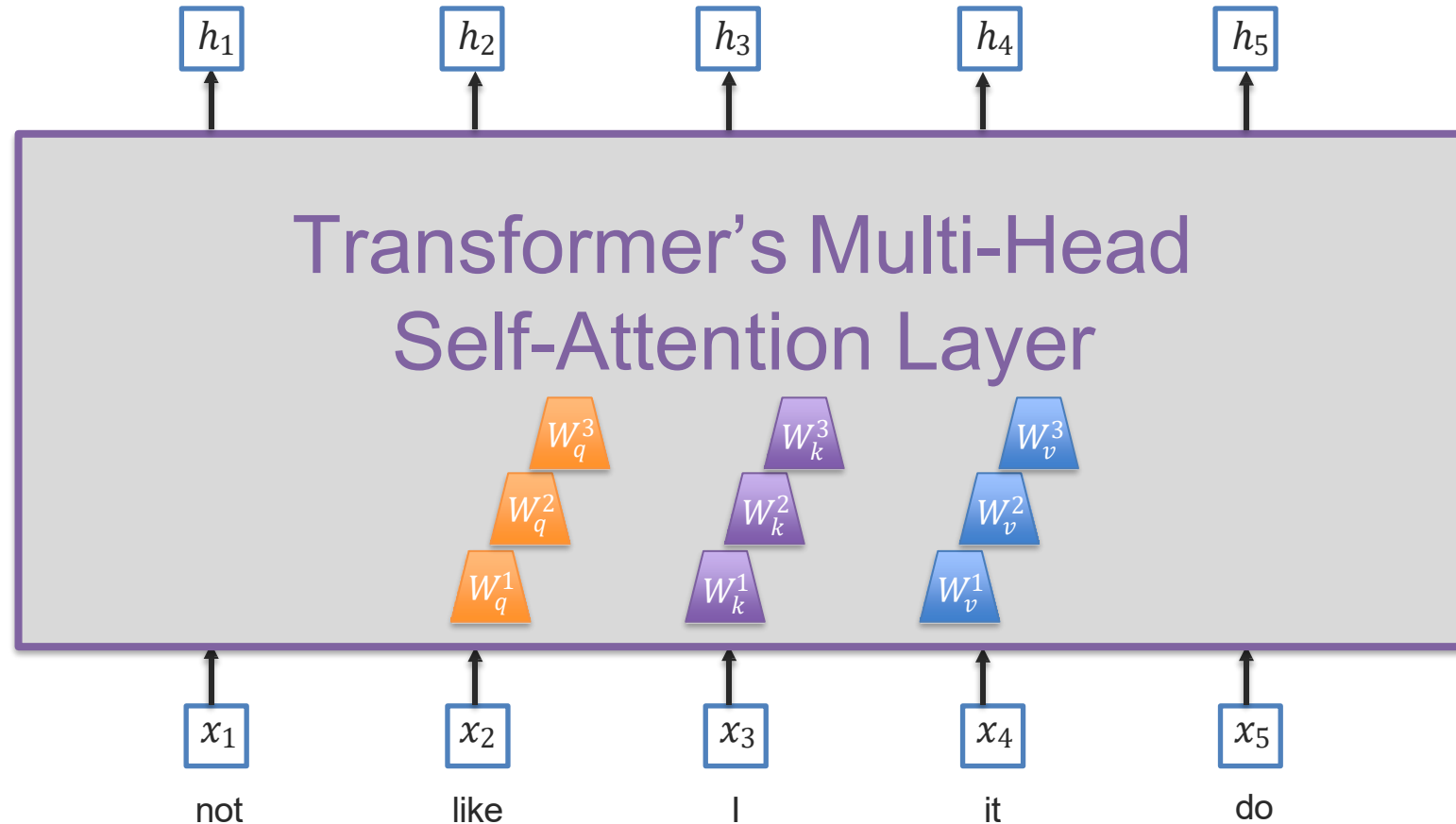
Transformer Self-Attention



Transformer Multi-Head Self-Attention



Transformer Multi-Head Self-Attention



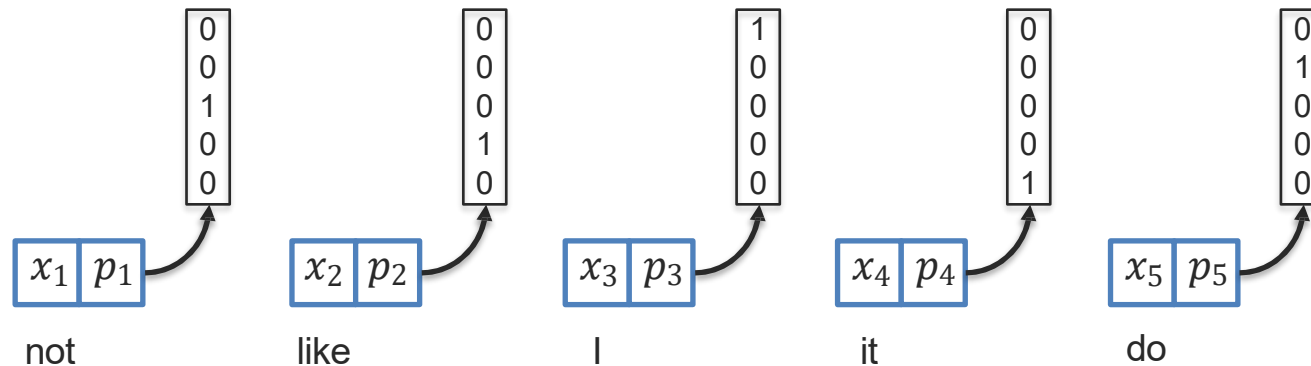
What happens if the words are shuffled?

Position embeddings

❑ Position information is not encoded in a self-attention module

How can we encode position information?

Simple approach: one-hot encoding

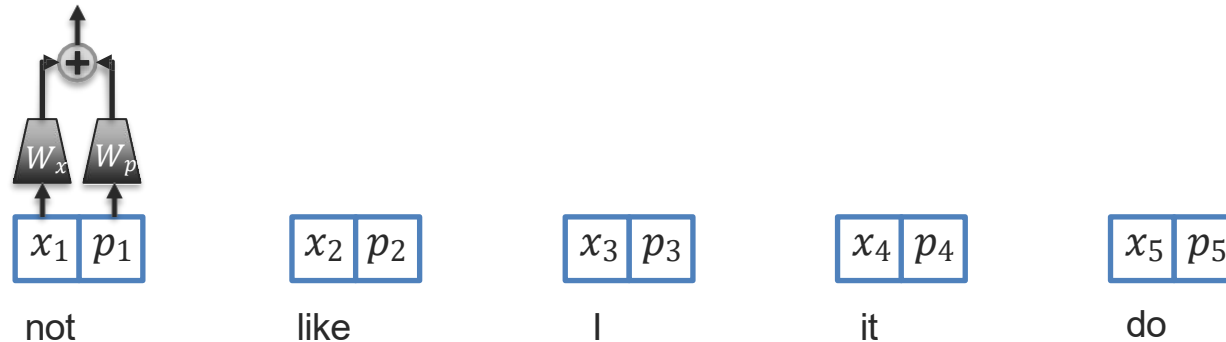


Position embeddings

- ❑ Position information is not encoded in a self-attention module

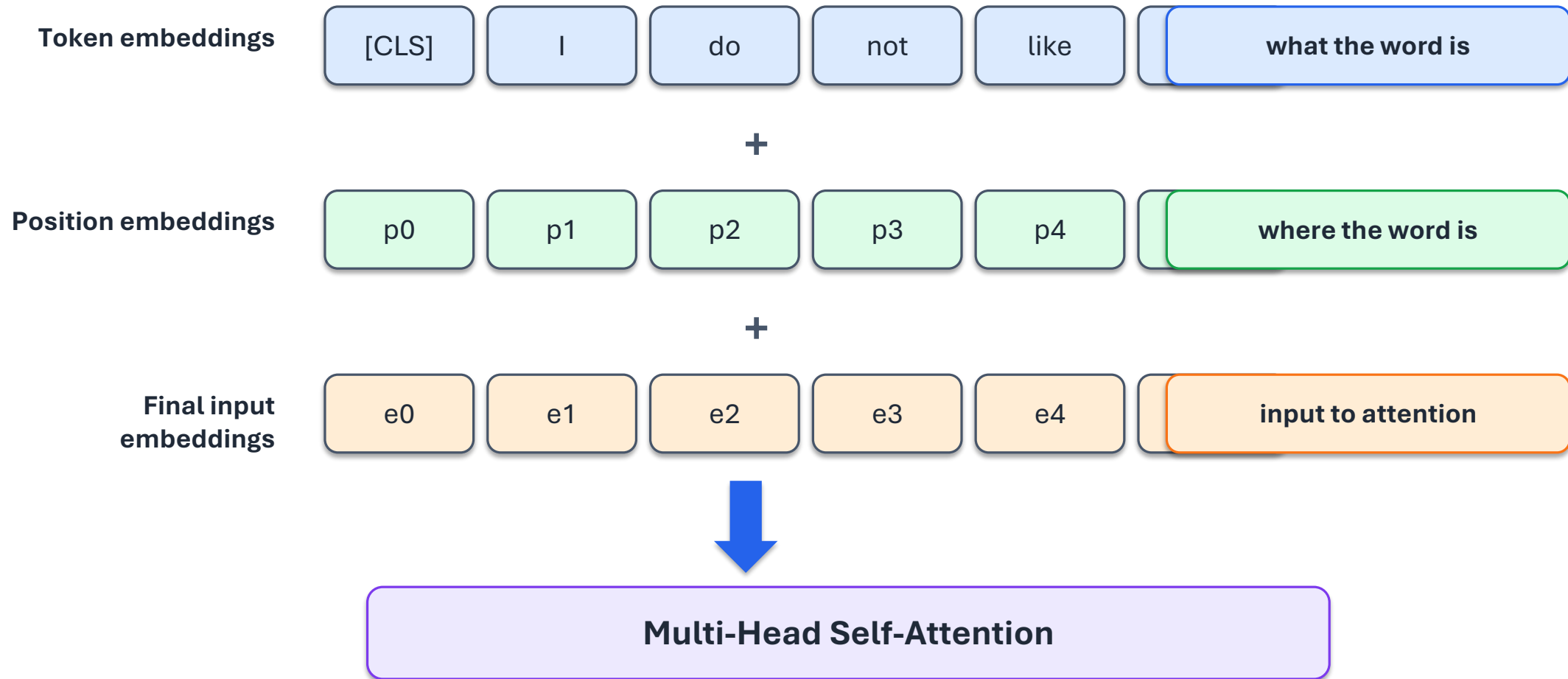
How can we encode position information?

Simple approach: one-hot encoding + linear embeddings + $\left\{ \begin{array}{l} \text{Sum} \\ \text{- or -} \\ \text{concat} \end{array} \right.$

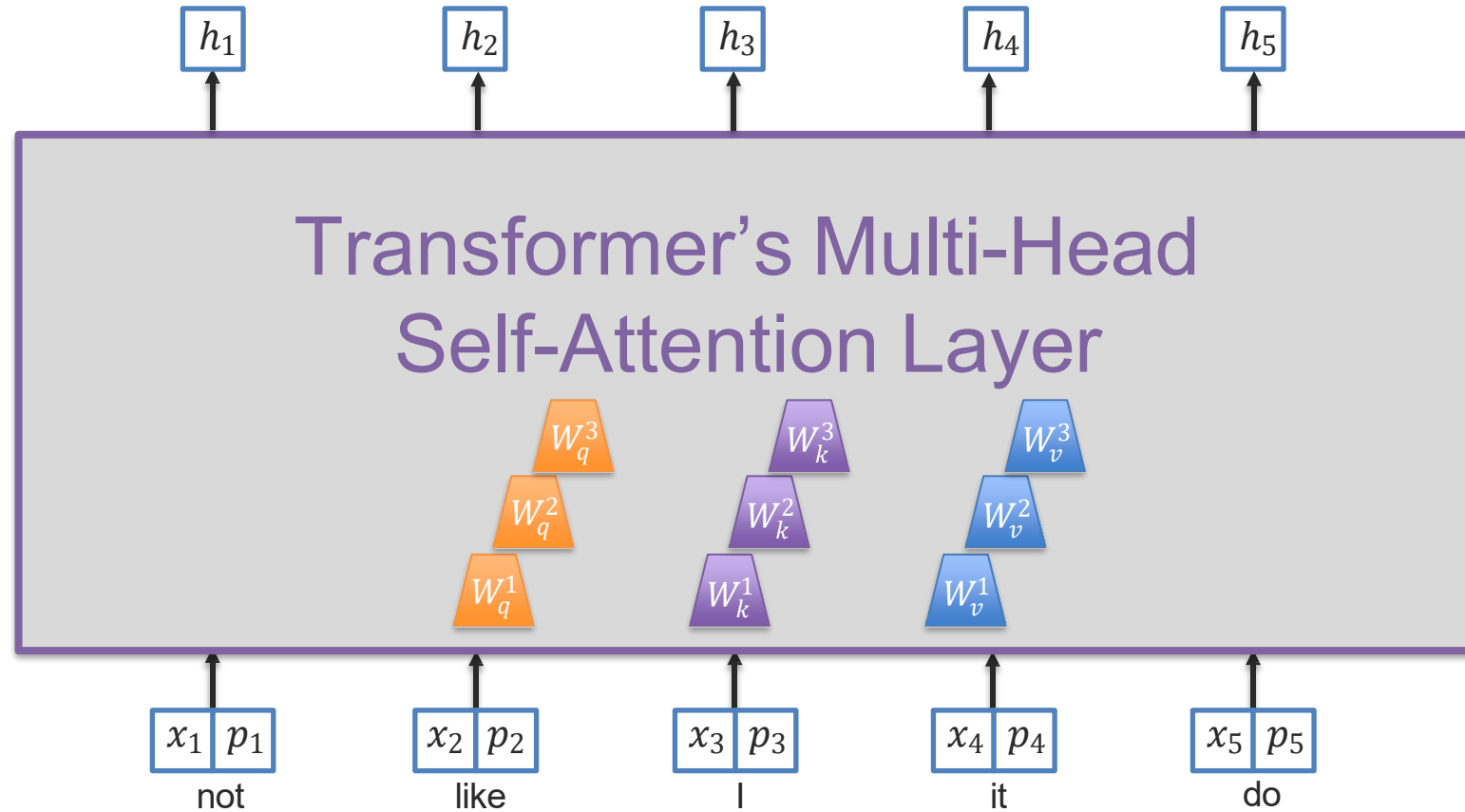


Token Embedding + Position Embedding

Transformer input = token embedding + position embedding

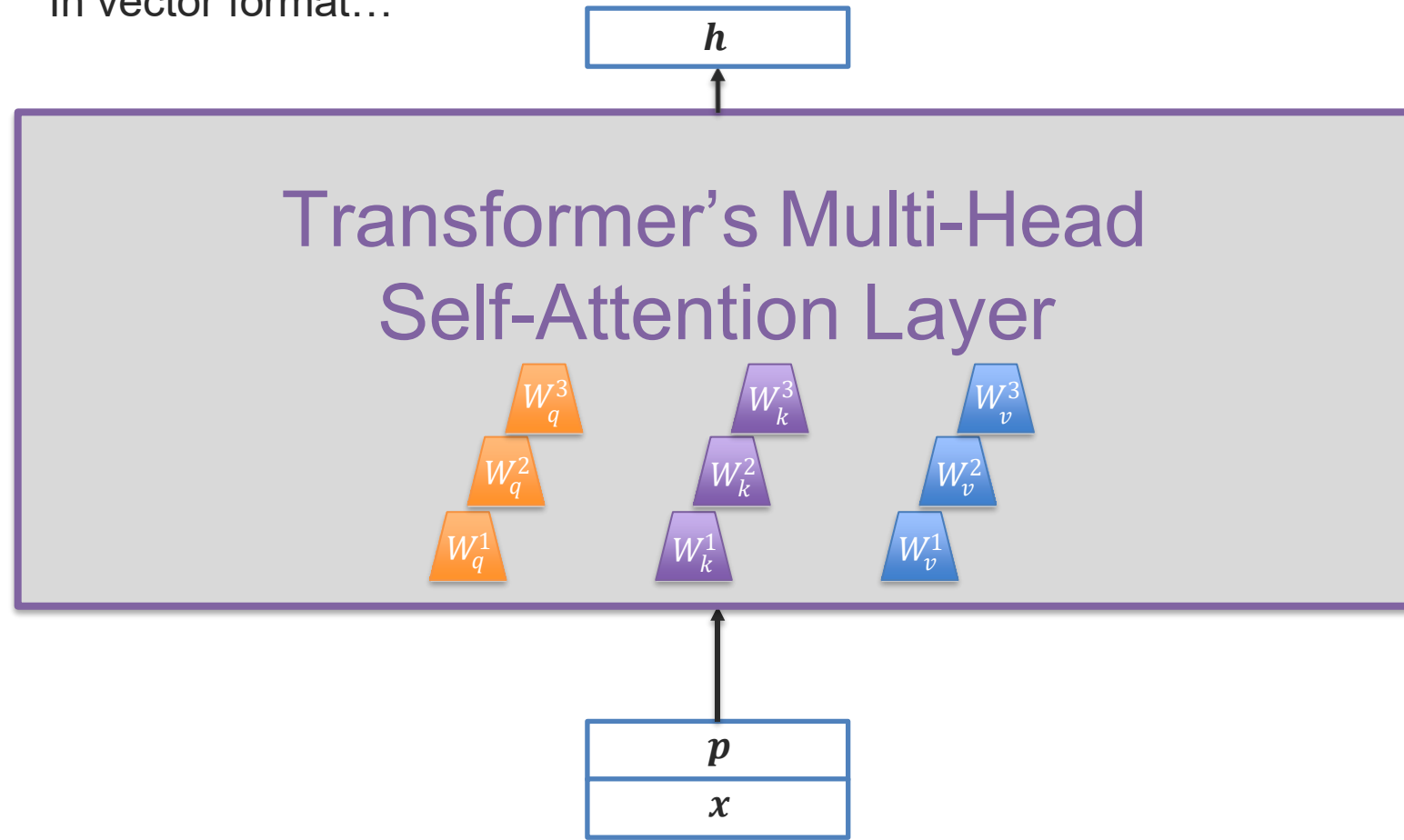


Transformer Multi-Head Self-Attention

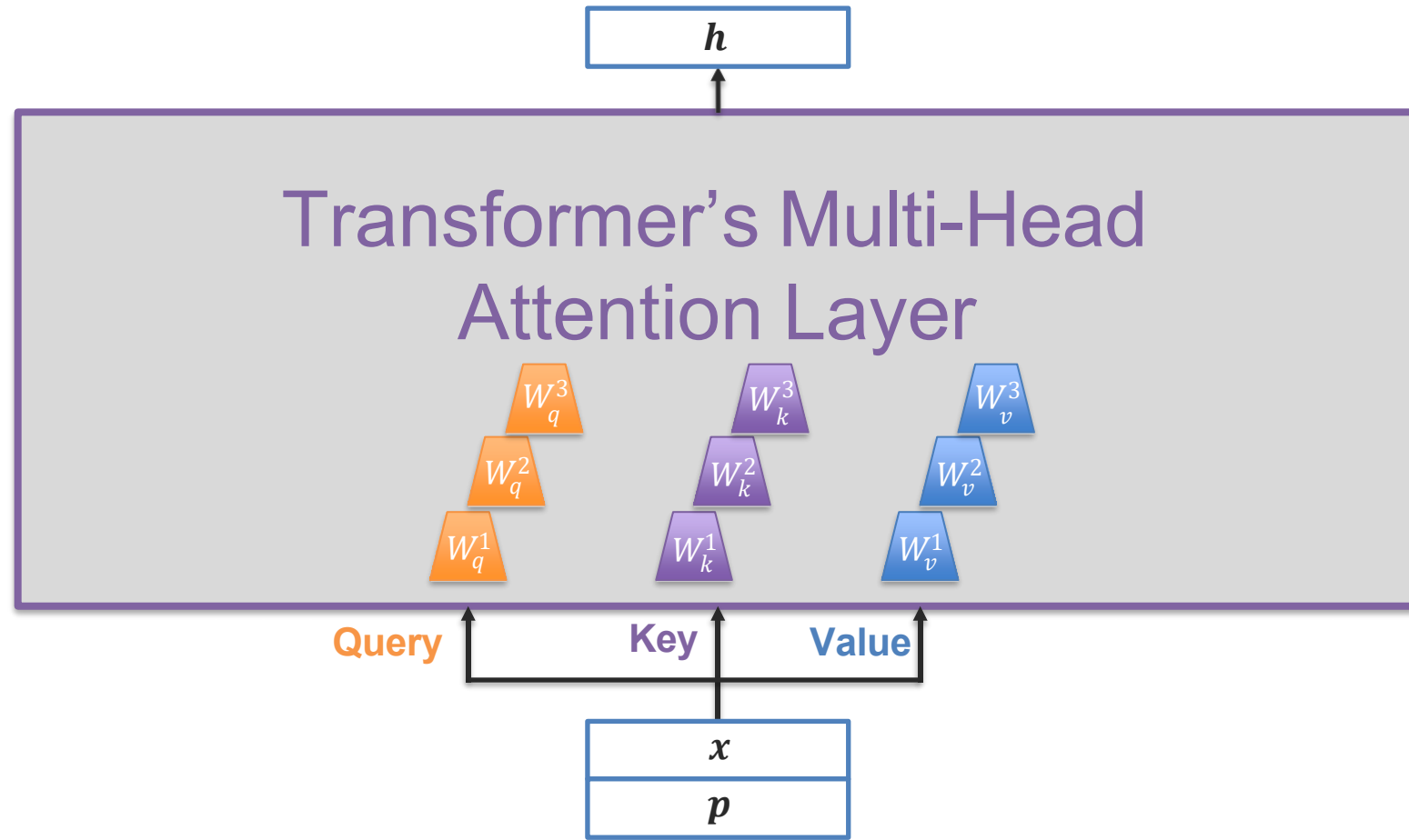


Transformer Multi-Head Self-Attention

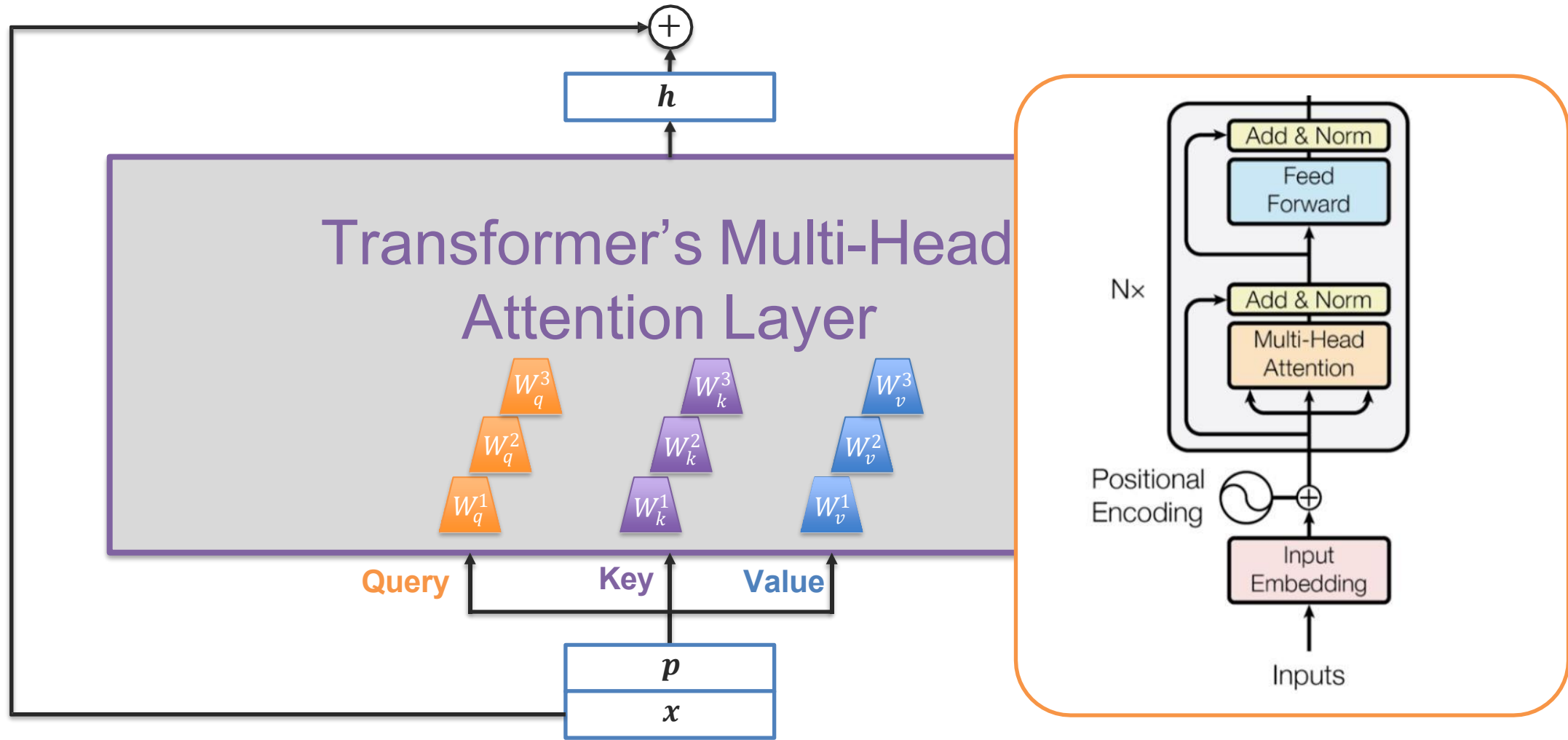
In vector format...



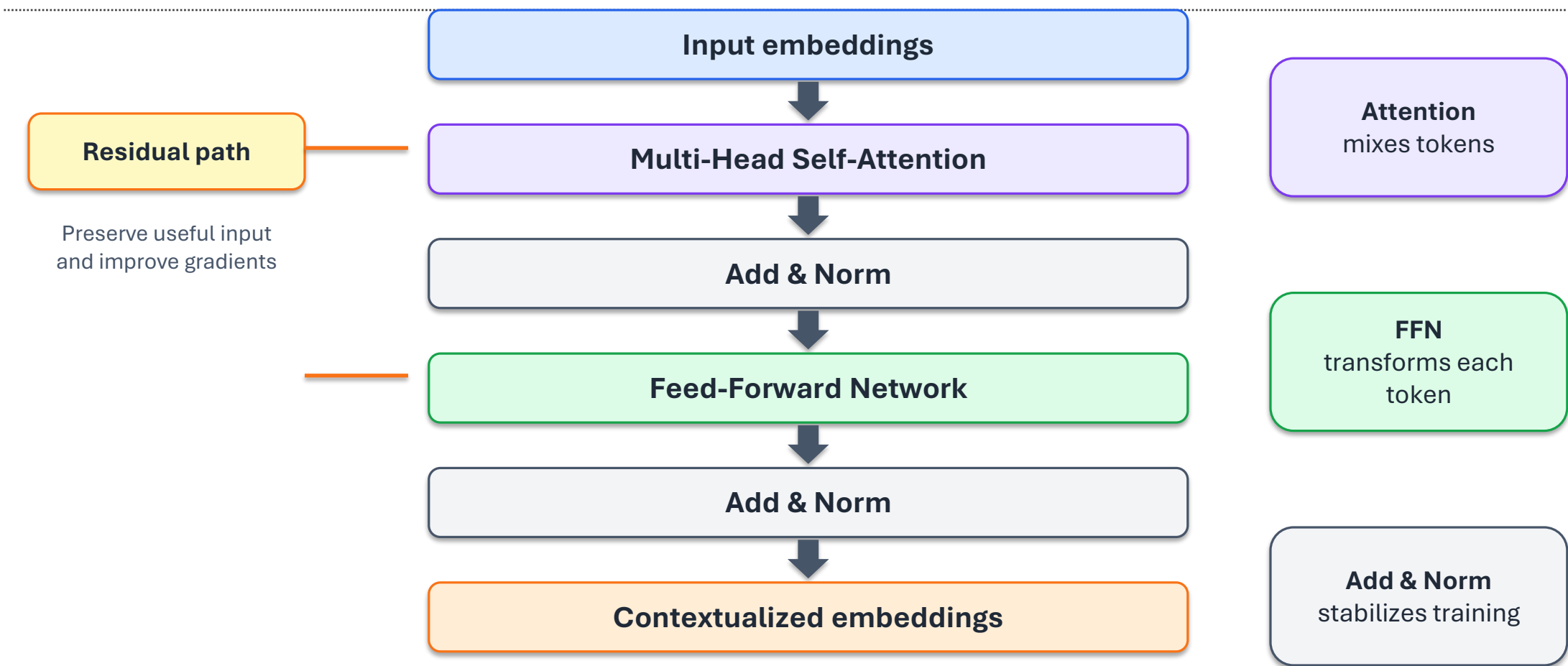
Transformer Multi-Head Attention



Transformer – Residual Connection



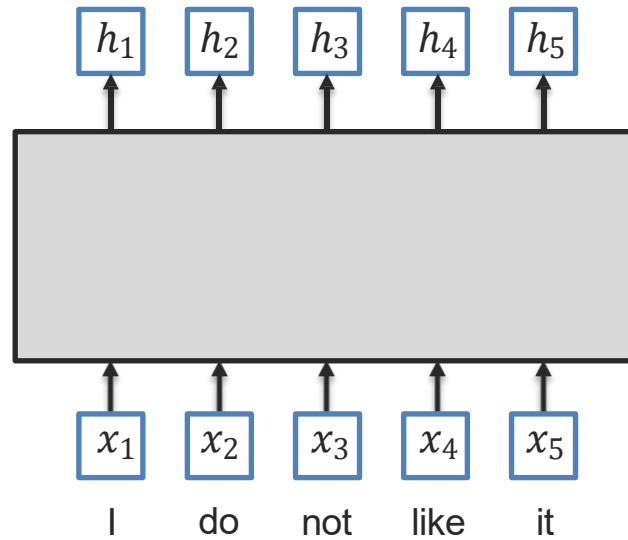
Inside One Transformer Encoder Block



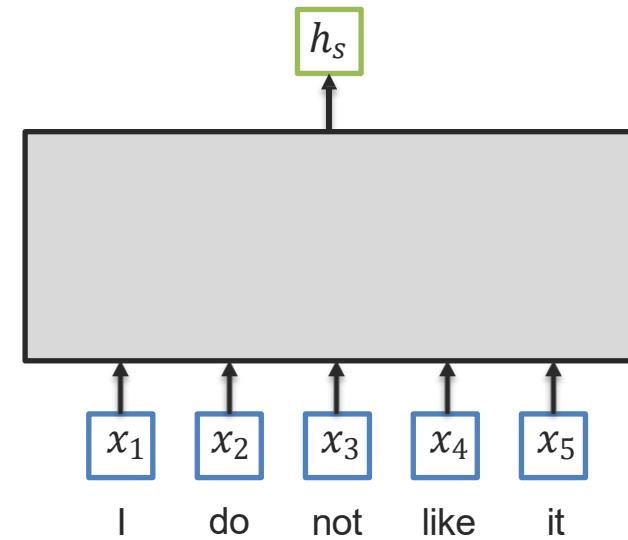
Language Pre-training

Token-level and Sentence-level Embeddings

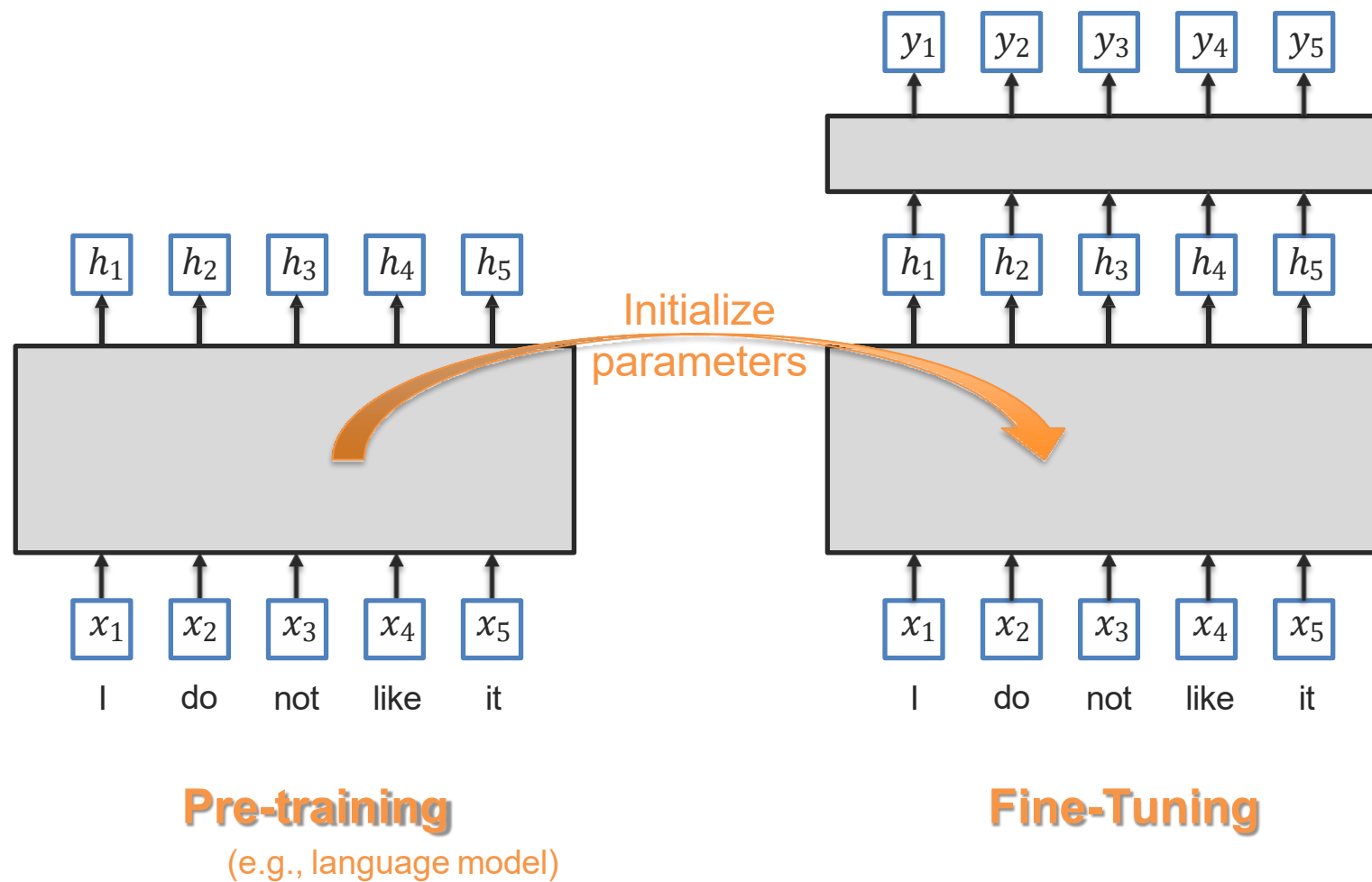
Token-level embeddings



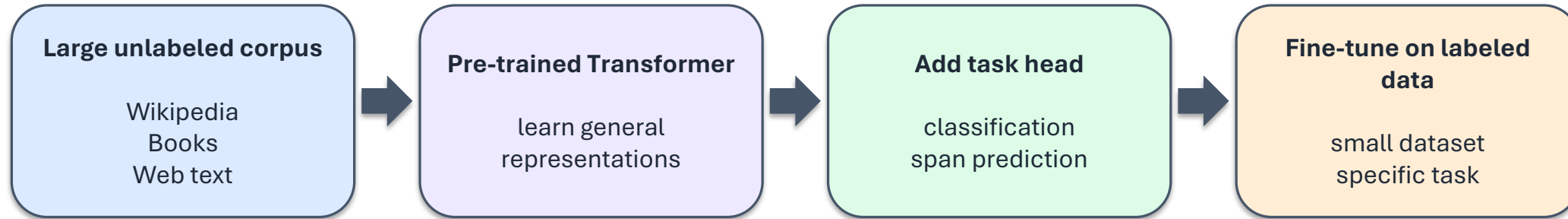
Sentence-level embedding



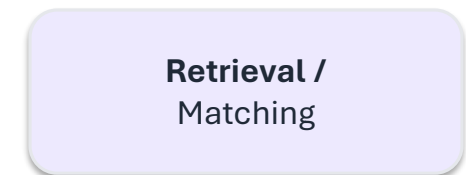
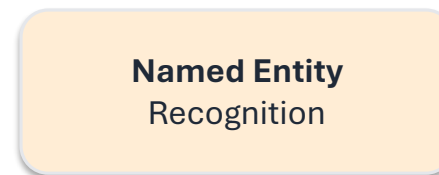
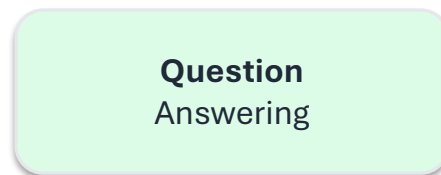
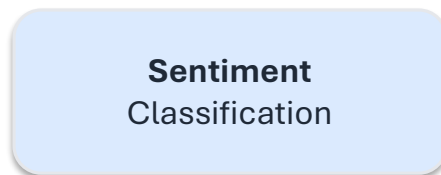
Pre-Training and Fine-Tuning



From Pre-training to Fine-tuning



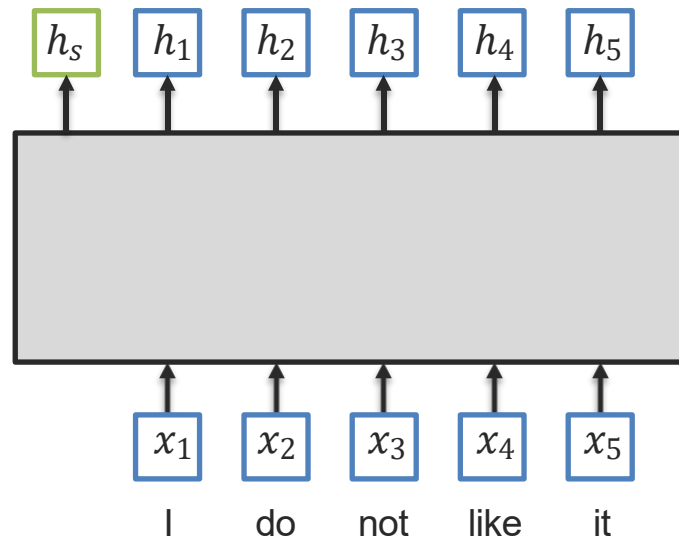
Same backbone, different task-specific heads



BERT: Bidirectional Encoder Representations from Transformers

Advantages:

- ① Jointly learn representation for token-level and sentence level
- ② Same network architecture for pre-training and fine-tuning

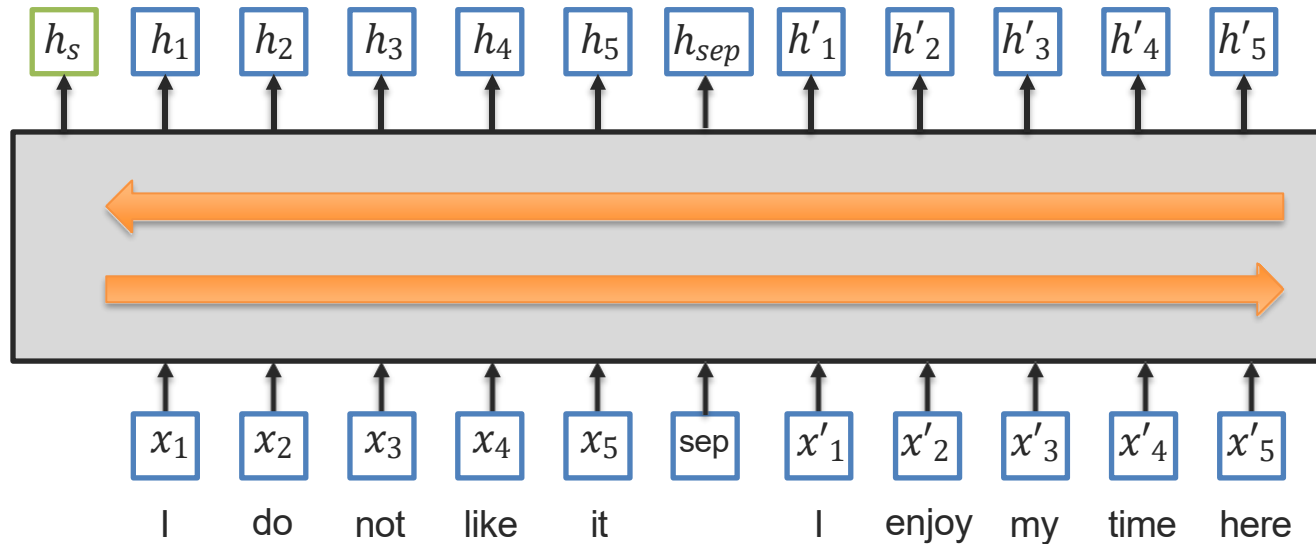


BERT: Bidirectional Encoder Representations from Transformers

Advantages:

- ① Jointly learn representation for token-level and sentence level
- ② Same network architecture for pre-training and fine-tuning
- ③ Can be used learn relationship between sentences
- ④ Models bidirectional and long-range interactions between tokens

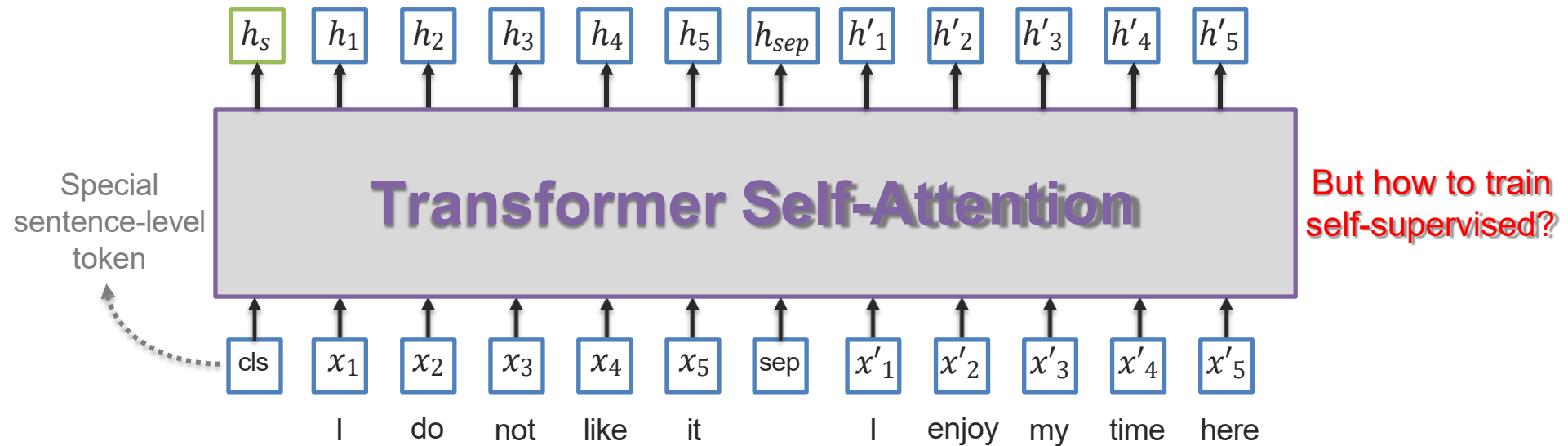
How can
we do all
this?



BERT: Bidirectional Encoder Representations from Transformers

Advantages:

- ① Jointly learn representation for token-level and sentence level
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- ④ Models bidirectional interactions between tokens



BERT Input Representation

$$\text{Input} = \text{token embedding} + \text{position embedding} + \text{segment embedding}$$

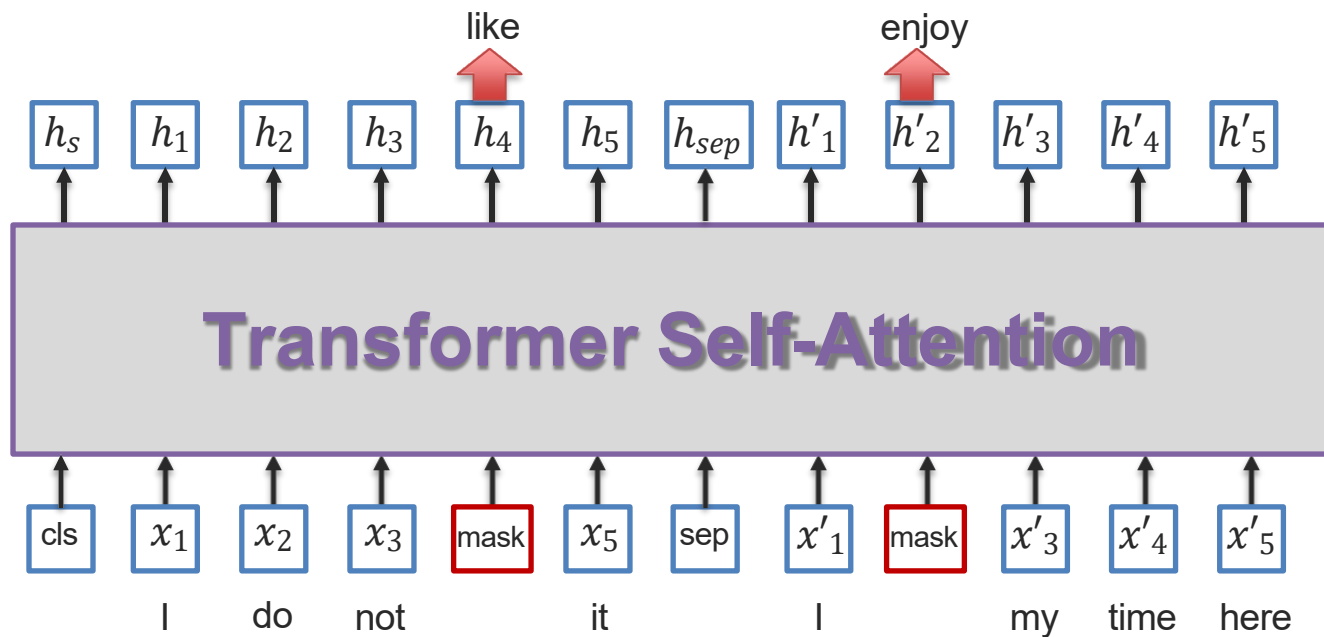


Pre-training BERT Model

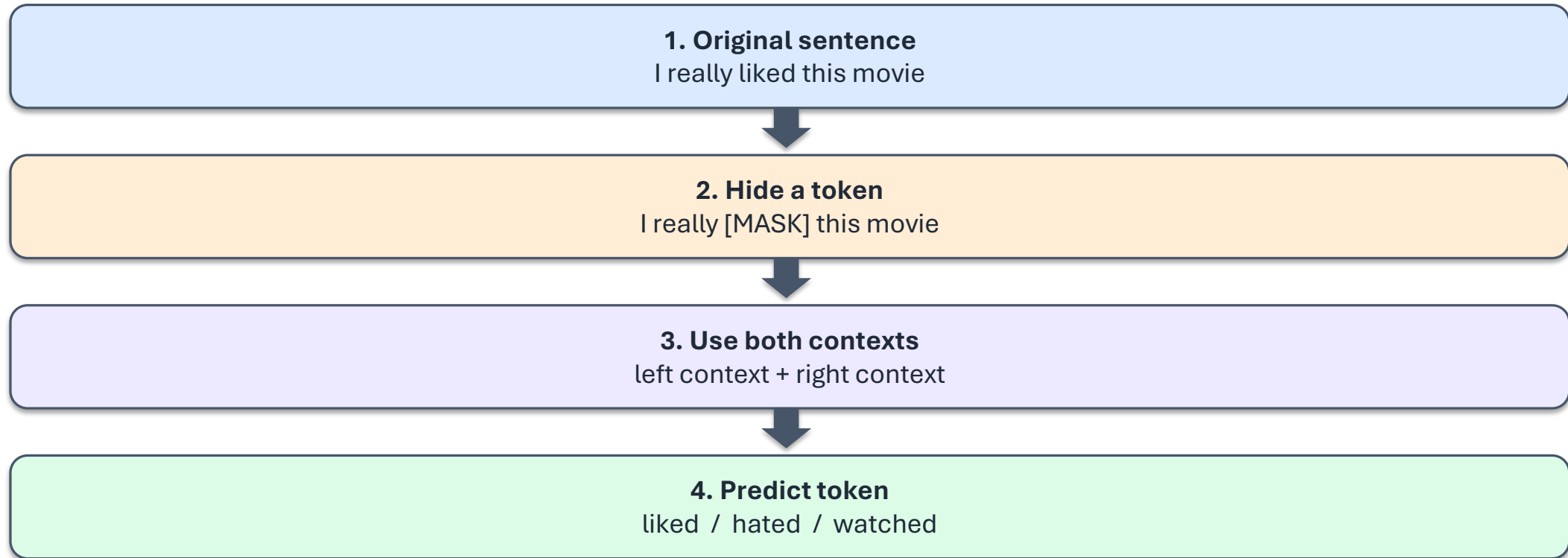
1 Masked Language Model

Randomly mask input tokens and then try to predict them

What is the loss function?



Masked Language Modeling Step by Step



Loss only on masked positions

Bidirectional context

Pre-training BERT Model

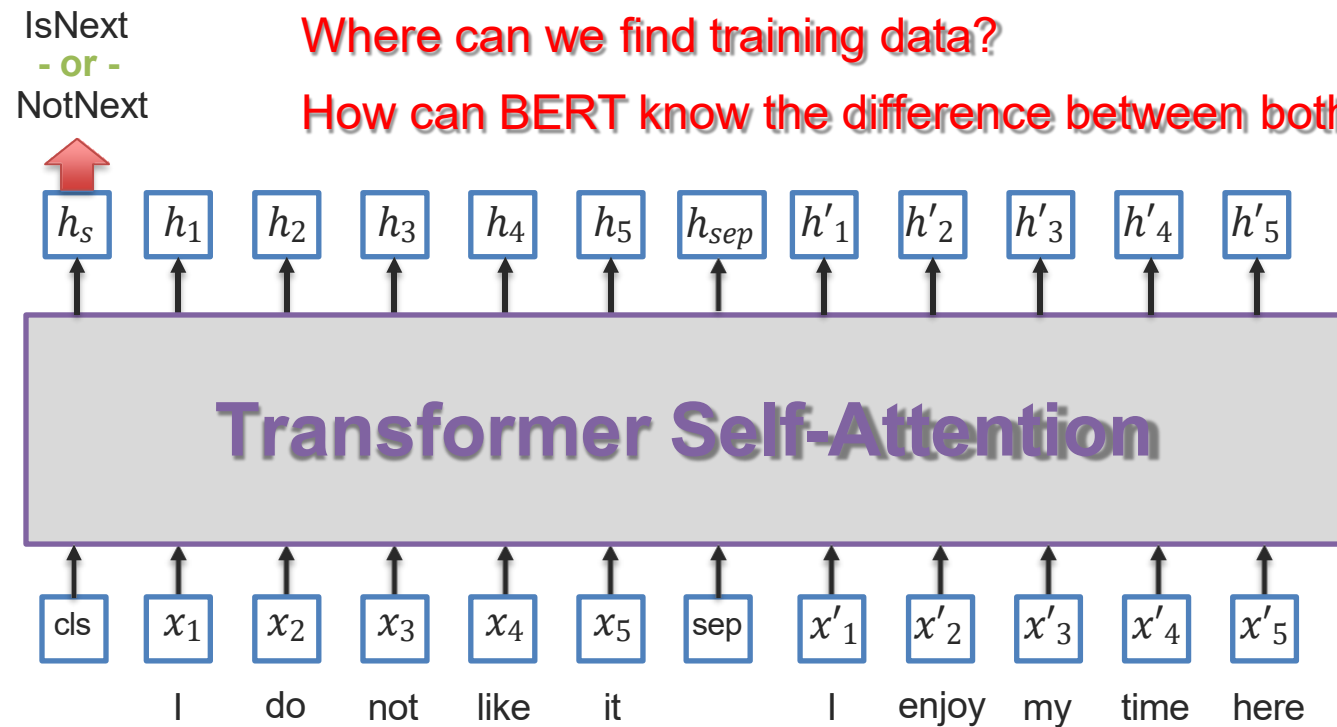
② Next Sentence Prediction

Given two sentences, predict if this is the next one or not

What is the loss function?

Where can we find training data?

How can BERT know the difference between both sentences?



Next Sentence Prediction Intuition

Task: predict whether sentence B follows sentence A

Example 1

A: I missed the bus.

B: I arrived late to class.

Output: IsNext

Example 2

A: I missed the bus.

B: The cake was delicious.

Output: NotNext

[CLS] sentence A [SEP] sentence B [SEP] → 2-way classifier

Fine-Tuning BERT

4 Question-answering: find start/end of the answer in the document

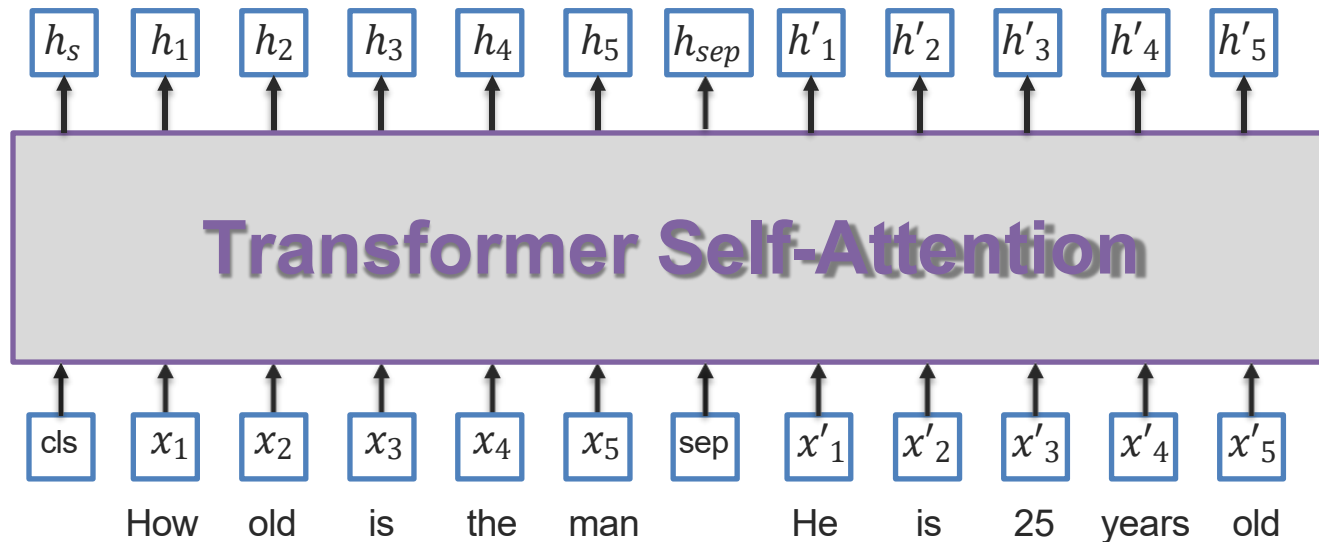
Paragraph: “... Other legislation followed, including the Migratory Bird Conservation Act of 1929, a 1937 treaty prohibiting the hunting of right and gray whales, and the Bald Eagle Protection Act of 1940. These later laws had a low cost to society—the species were relatively rare—and little opposition was raised.”

Question 1: “Which laws faced significant *opposition*?”

Plausible Answer: *later laws*

Question 2: “What was the name of the 1937 *treaty*?”

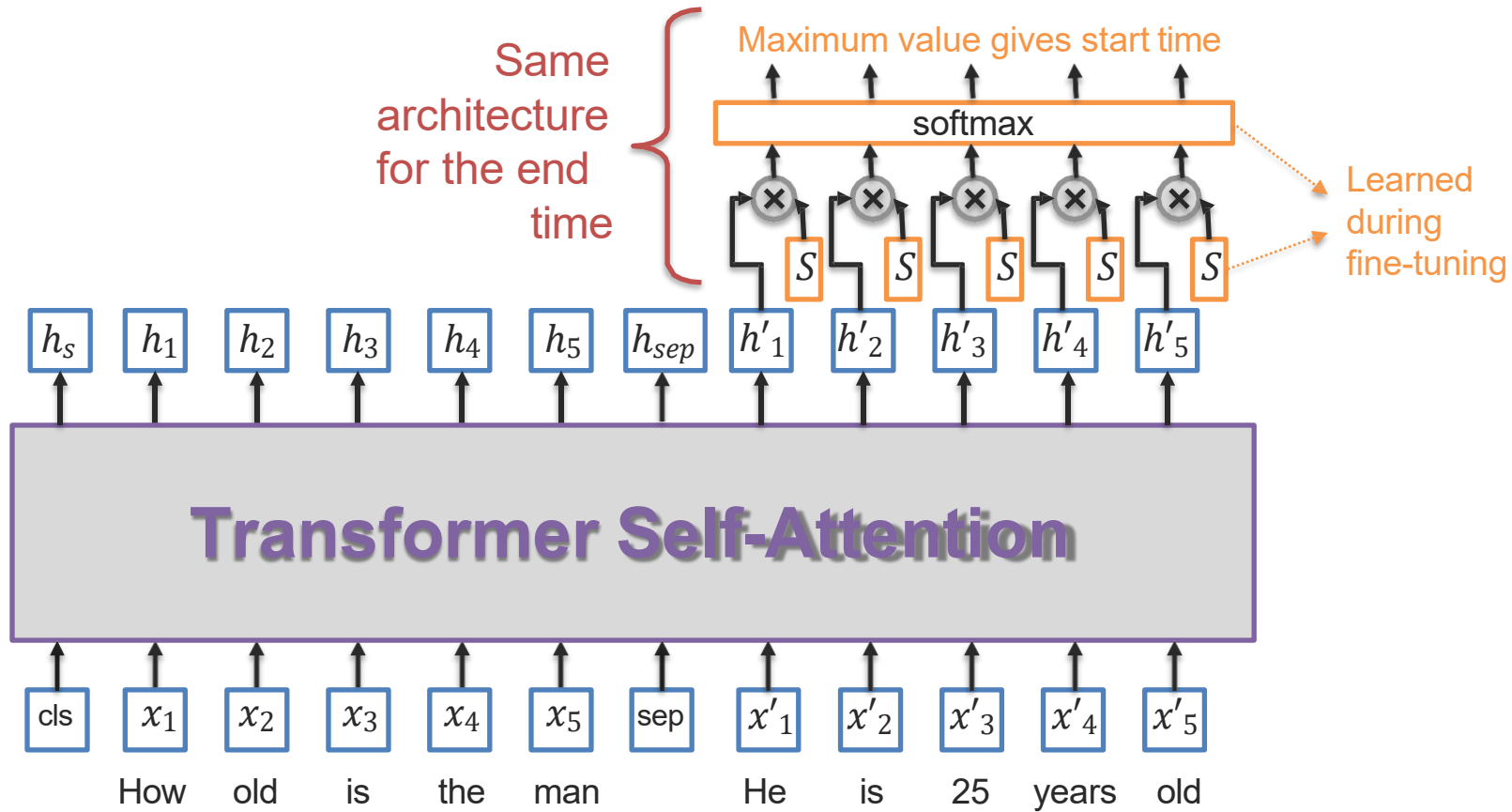
Plausible Answer: *Bald Eagle Protection Act*



How?

Fine-Tuning BERT

- 4 Question-answering: find start/end of the answer in the document



Question Answering with BERT

Input: question + passage

Question:

How old is the man?

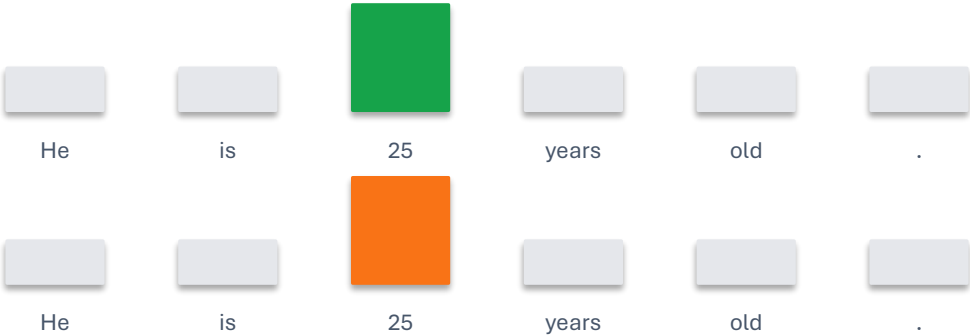
Start probability

Passage:

He is 25 years old .

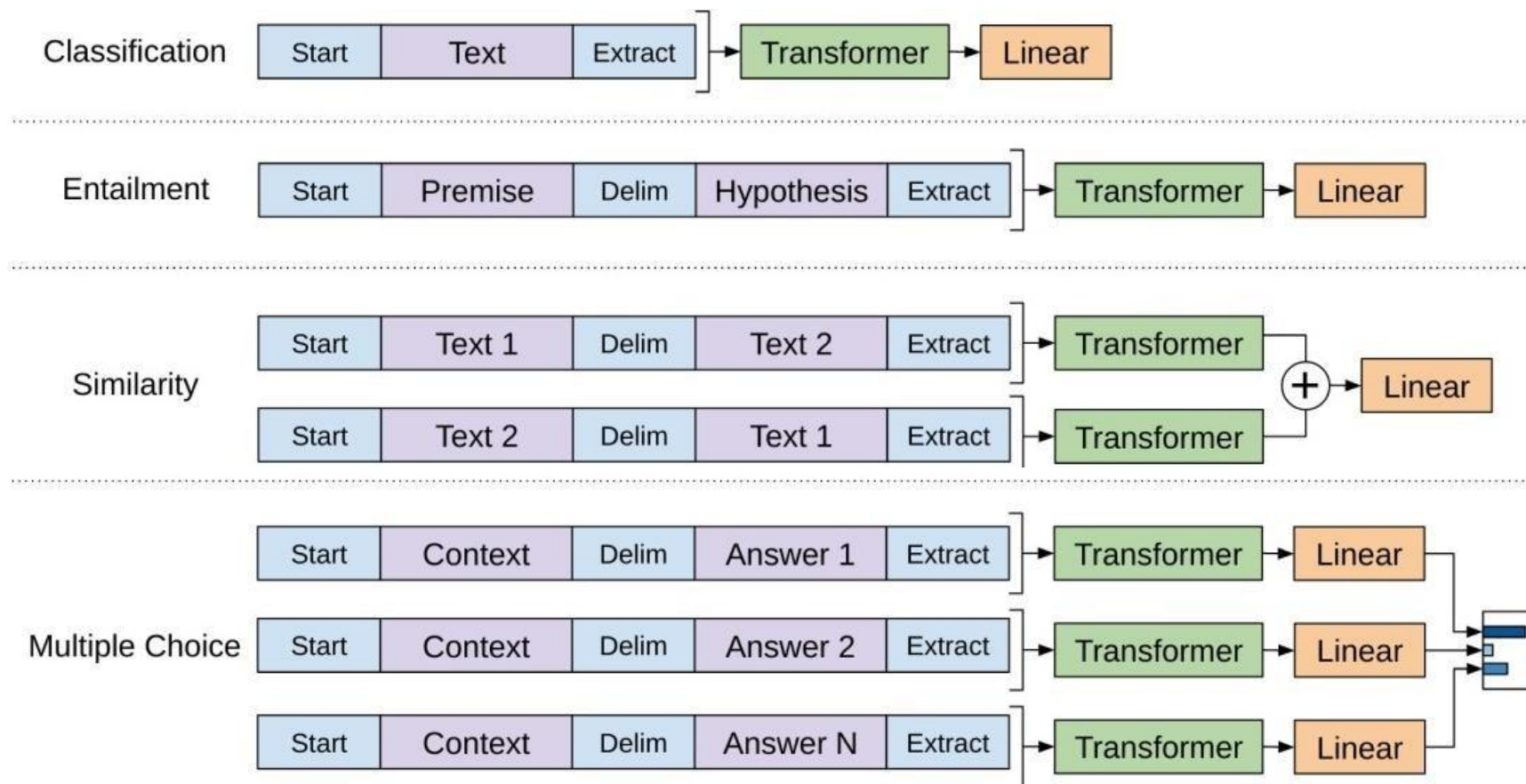
End probability

Answer span



BERT predicts two positions:
start token and end token

Other Fine-tuning Approaches



Language-Vision Transformers

What Is Multimodal Data?

Multimodal learning combines complementary signals

T

Text

words
sentences

IMG

Vision

images
regions



Audio

speech
tone

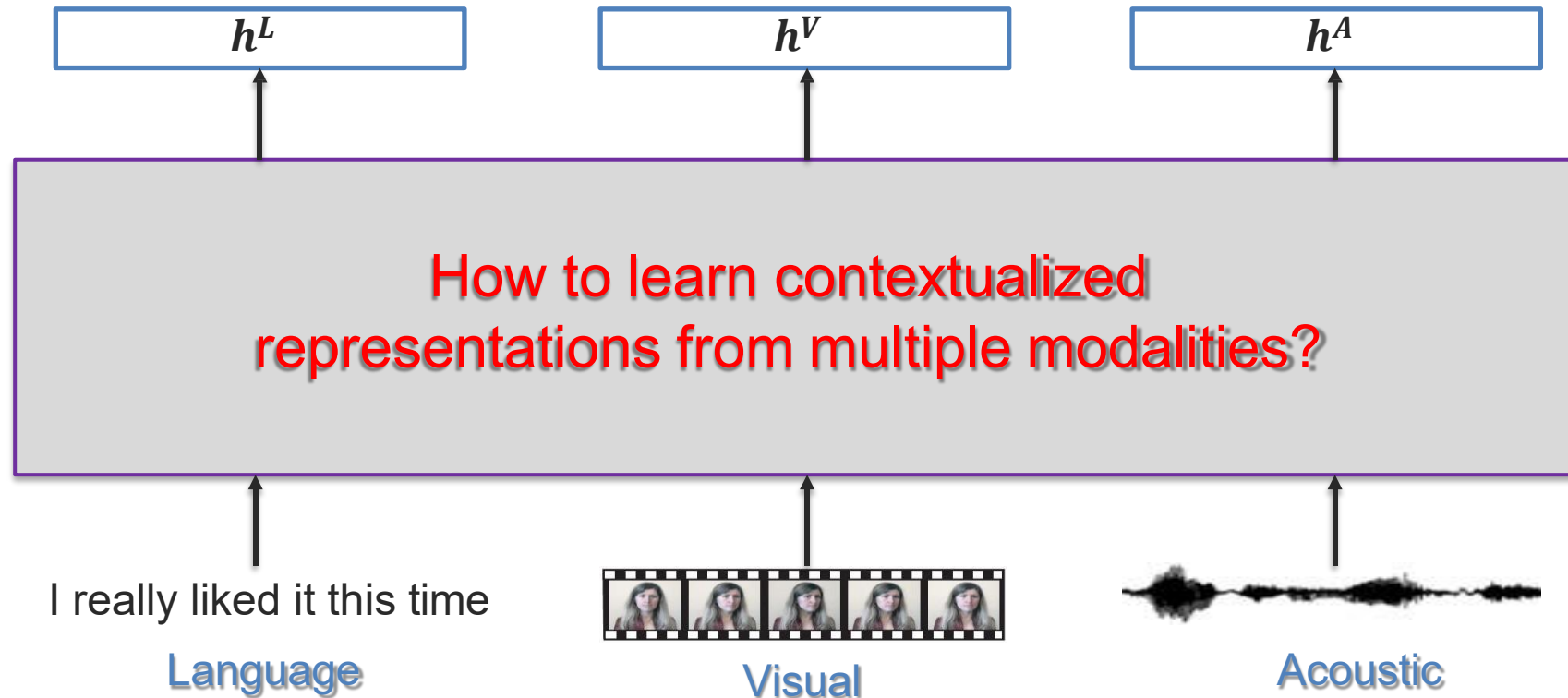


Video

frame
sequence

Goal: learn representations that use information from multiple modalities together.

Multimodal Embeddings



Option 1: Concatenate modalities and learn BERT transformer

Why Multimodal Learning Is Difficult

Feature mismatch

text tokens $\neq$ image regions
 $\neq$ audio frames

Length mismatch

different number of
elements per modality

Alignment problem

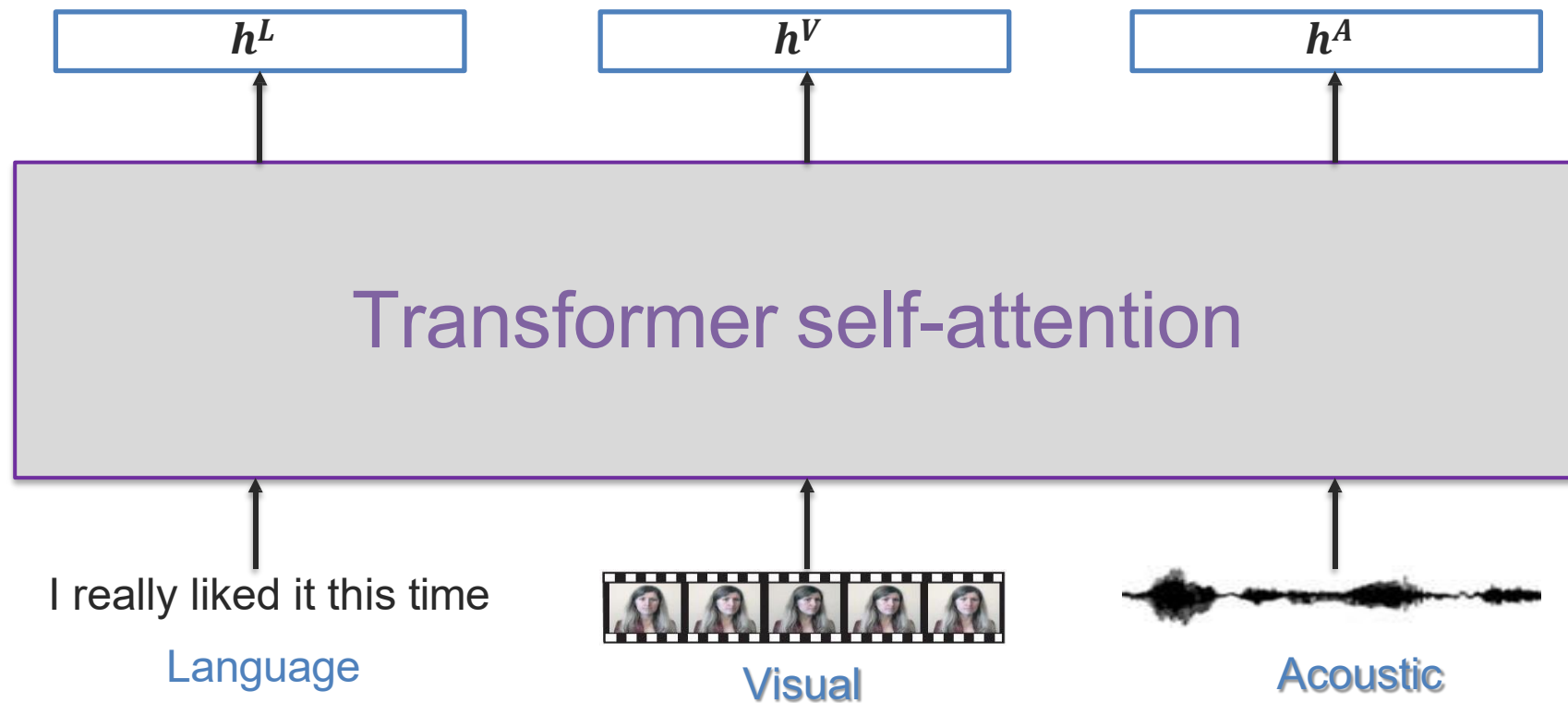
words, frames, and sounds
may not line up

Noise / missing data

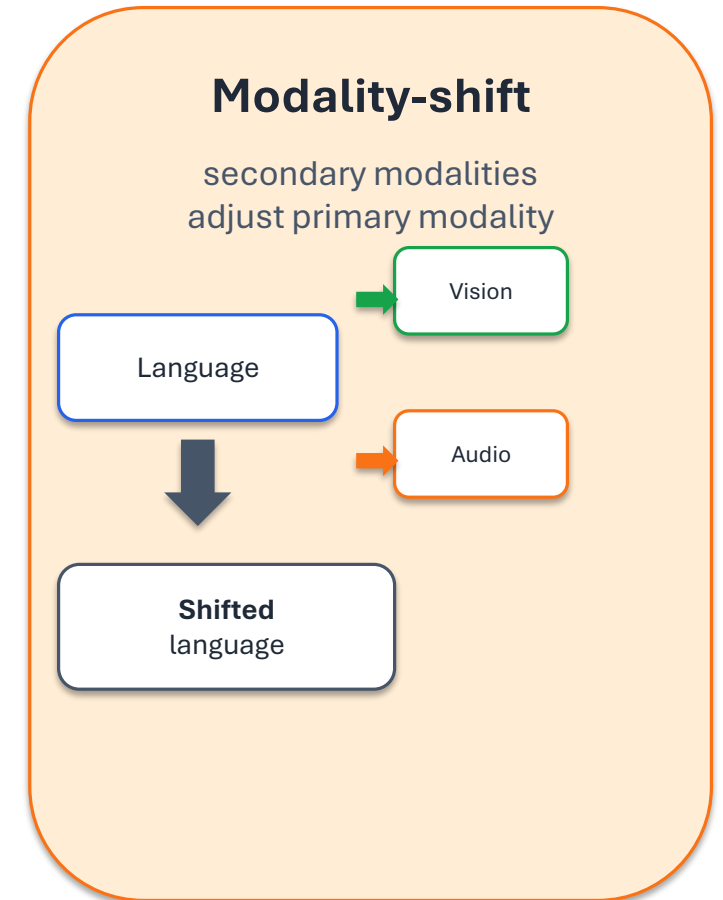
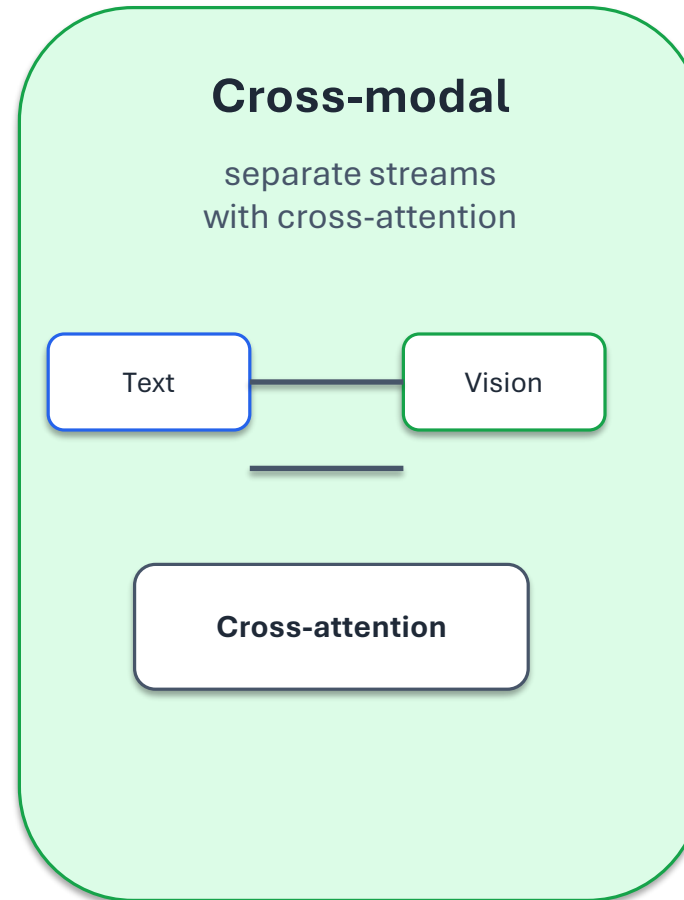
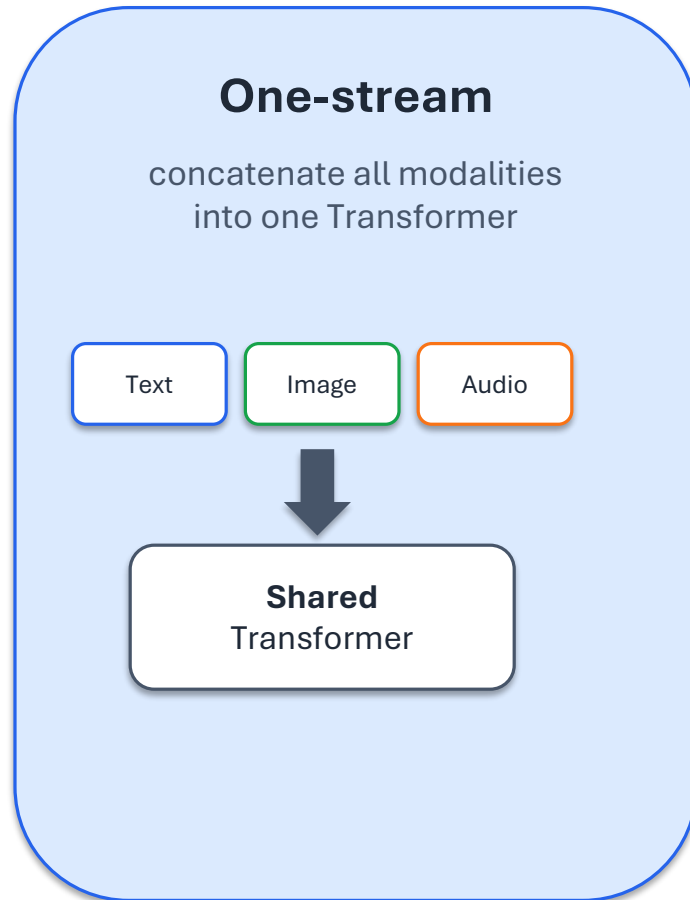
camera blur, silence,
incomplete modalities

The model must align, filter, and fuse information across modalities.

Simple Solution: Contextualized Multimodal Embeddings



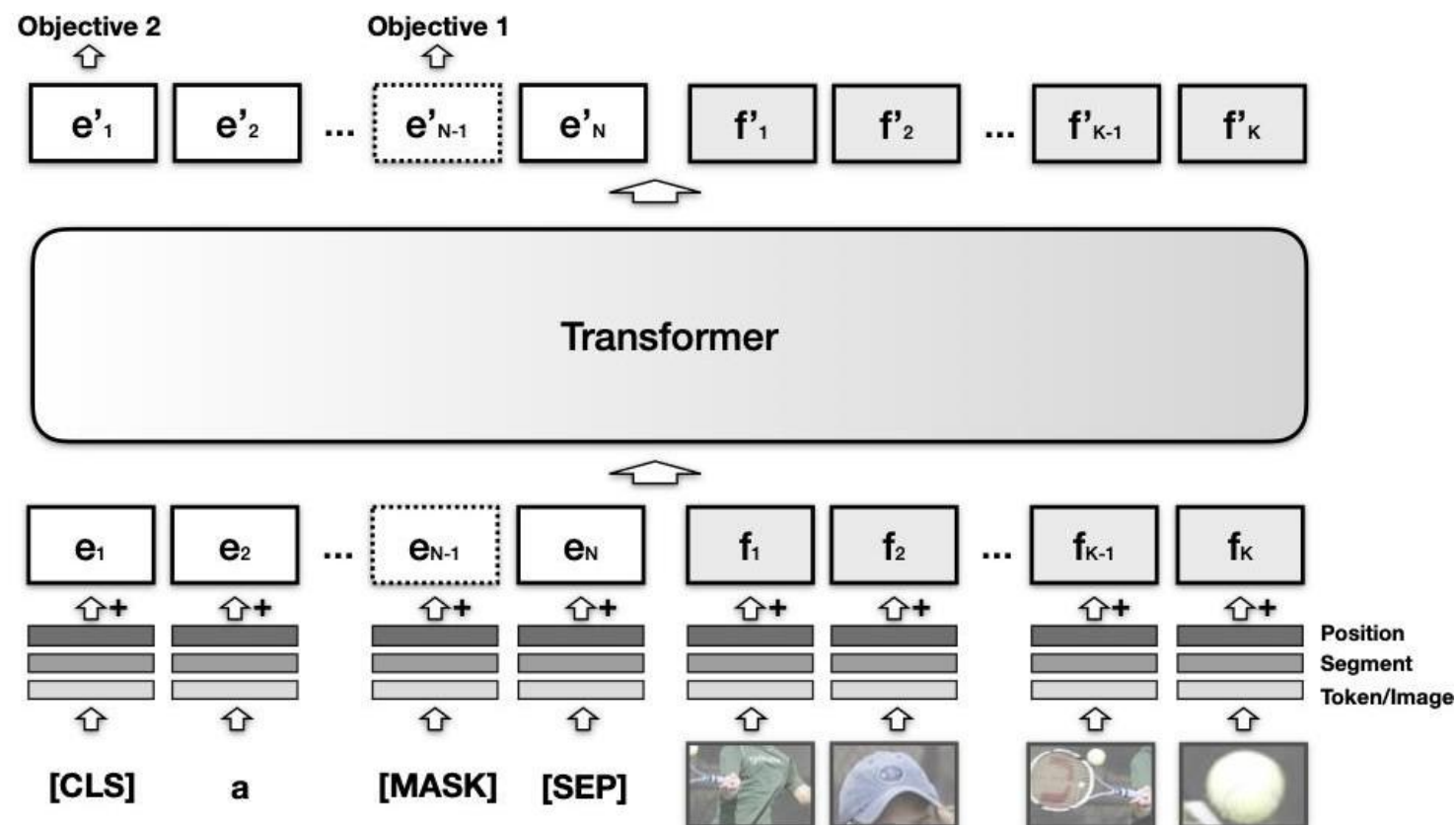
Three Design Patterns for Multimodal Transformers



VisualBERT



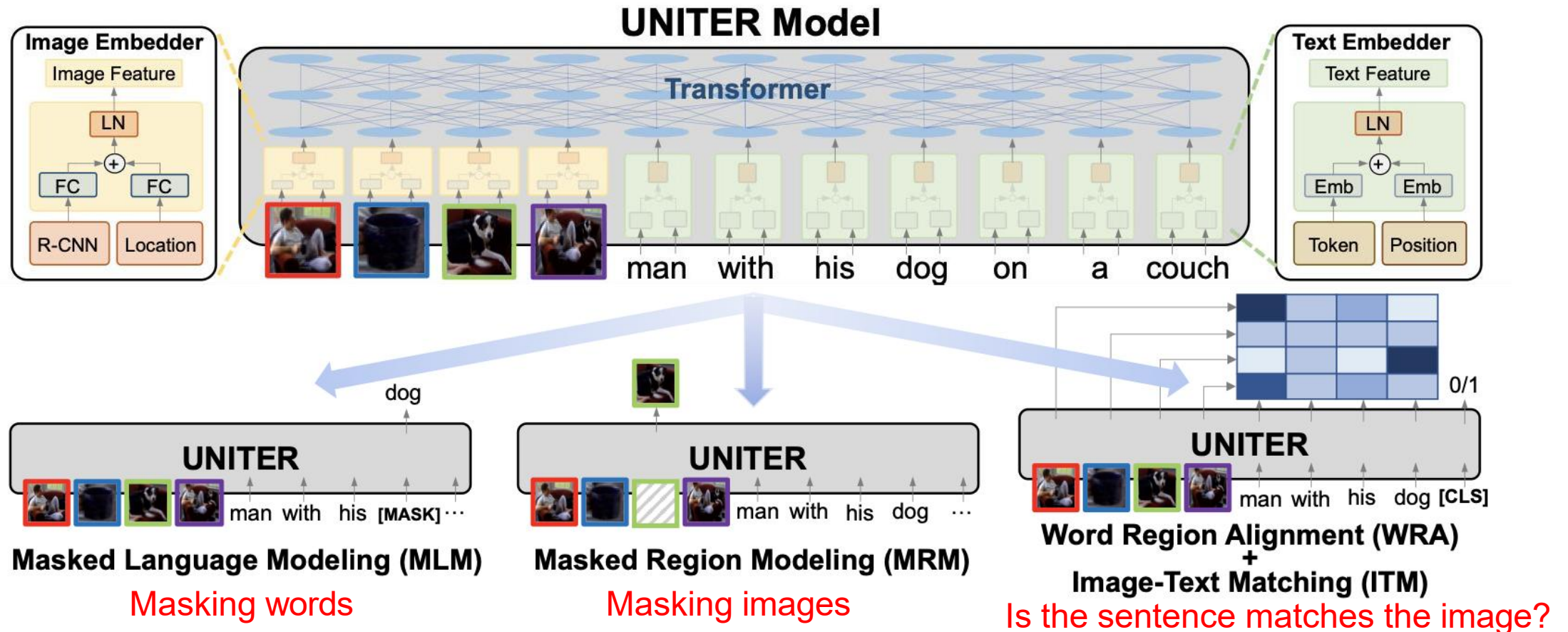
A person hits a ball with a tennis racket



Li, Liunian Harold, et al. "Visualbert: A simple and performant baseline for vision and language." *arXiv* (2019).

UNITER

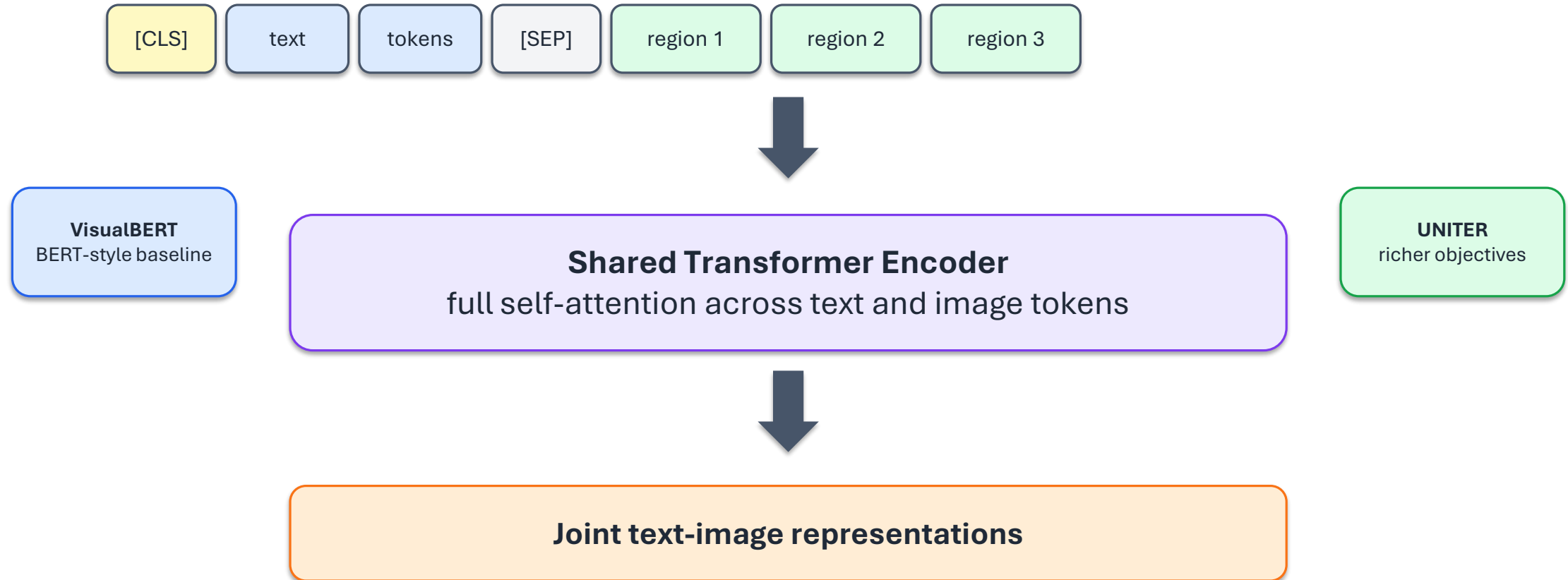
Similar Transformer architecture to BERT and VisualBERT... but with slightly different optimization



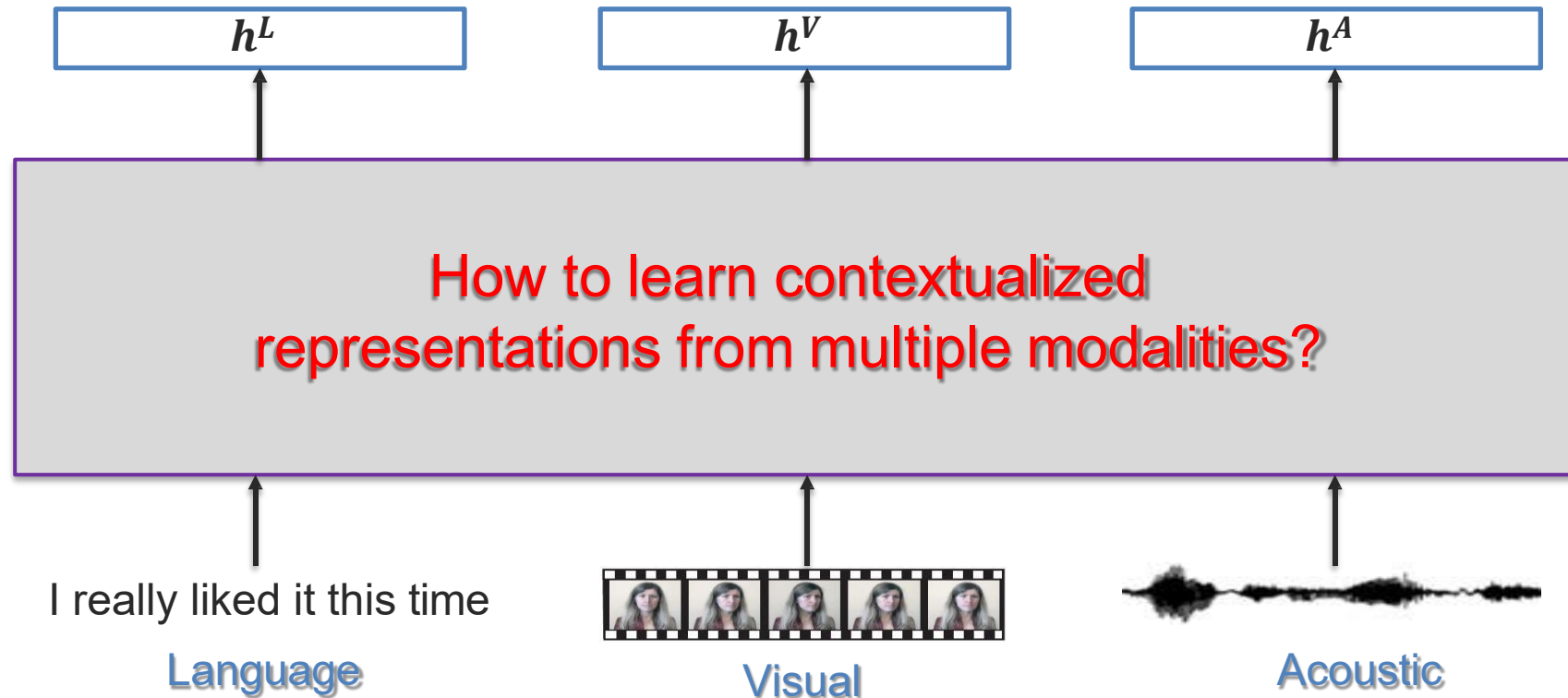
Chen, Yen-Chun, et al. "Uniter: Universal image-text representation learning." *European conference on computer vision*. 2020.

VisualBERT and UNITER: One-Stream Encoders

Text tokens and image regions become one joint sequence

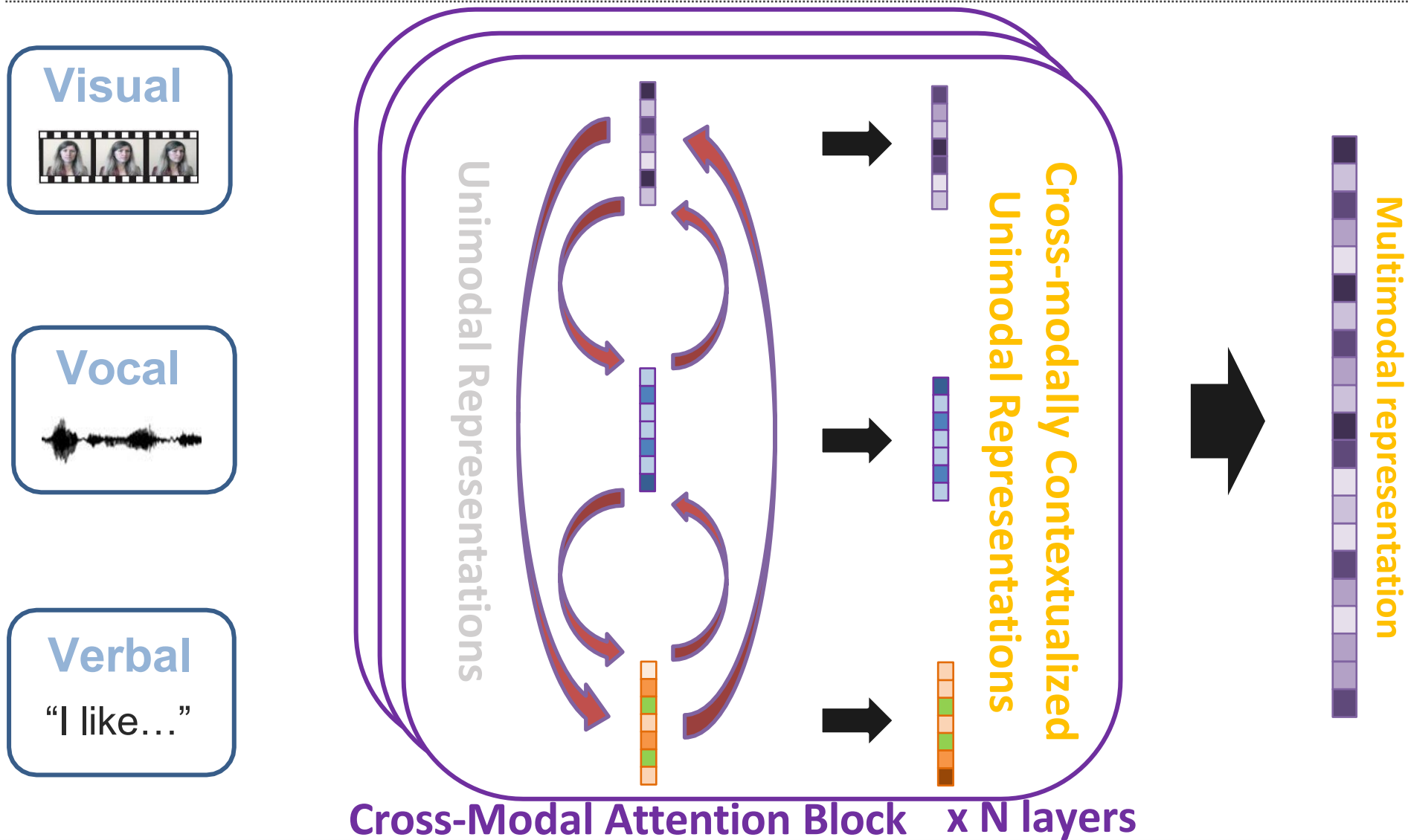


Multimodal Embeddings

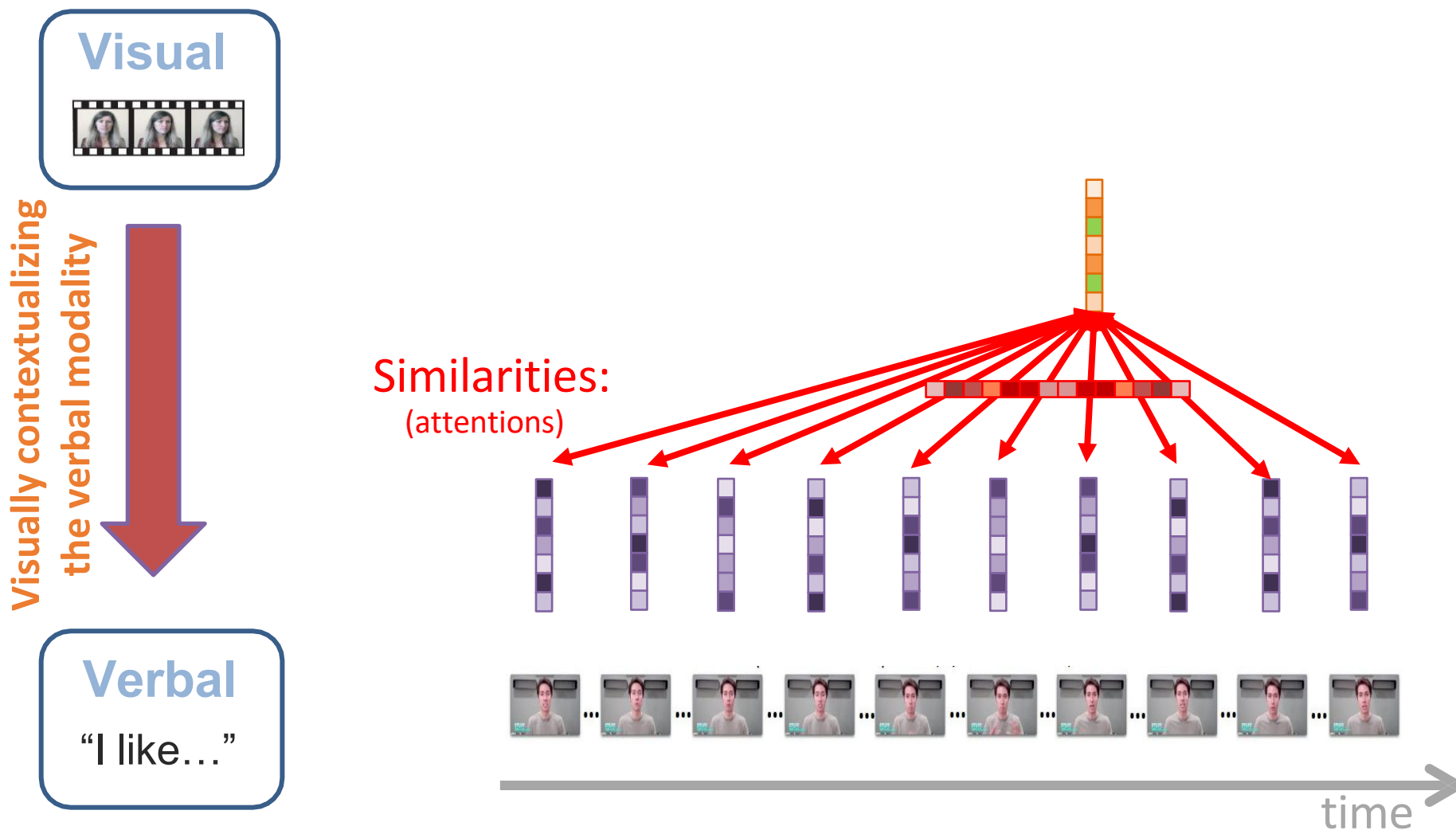


Option 2: Look at pairwise interactions between modalities

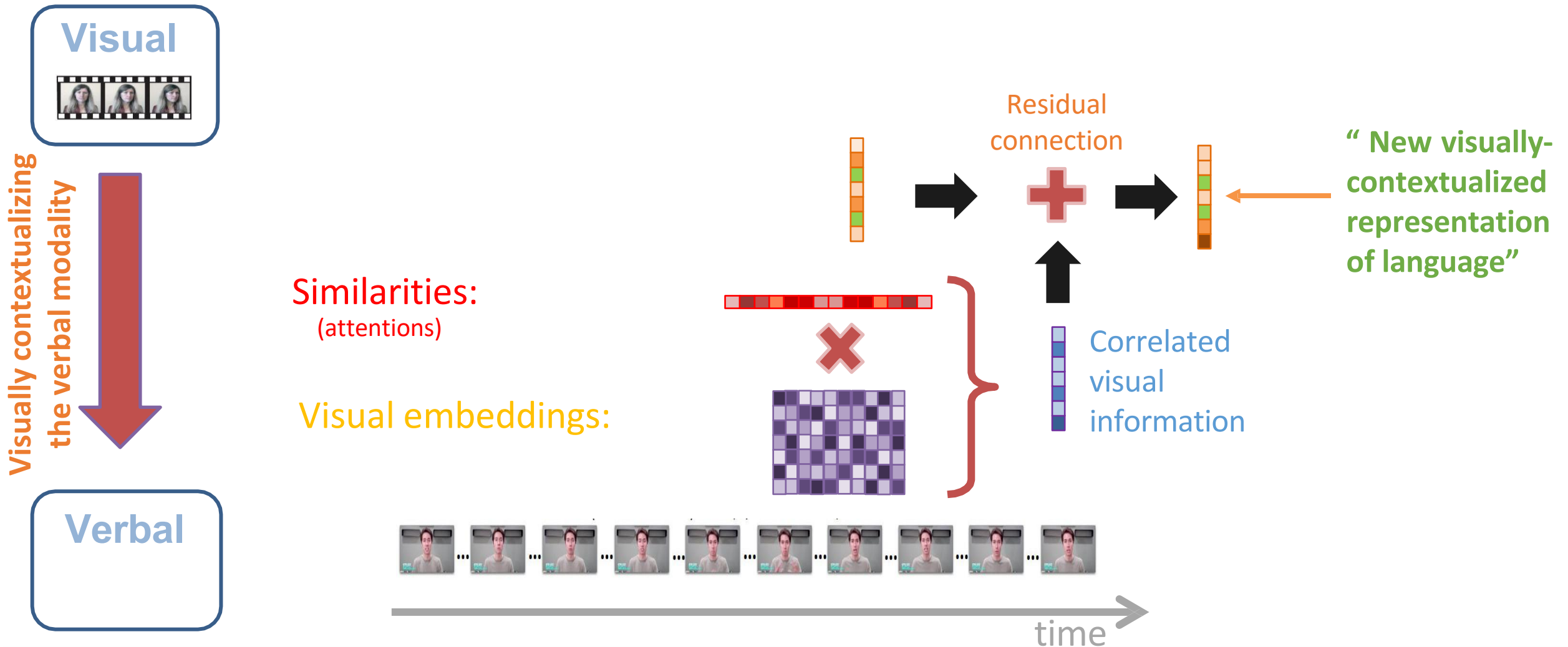
Multimodal Transformer – Pairwise Cross-Modal



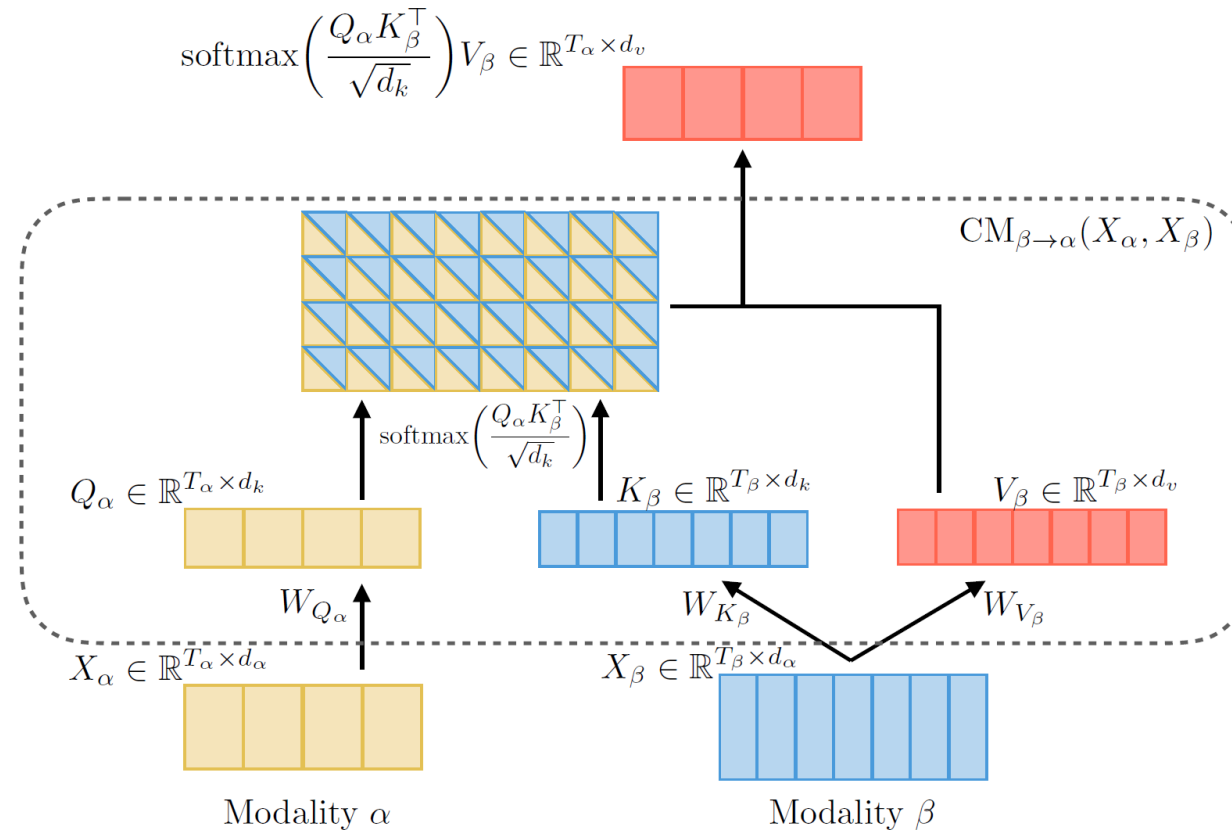
Cross-Modal Transformer Module ($V \rightarrow L$)



Cross-Modal Transformer Module ($V \rightarrow L$)

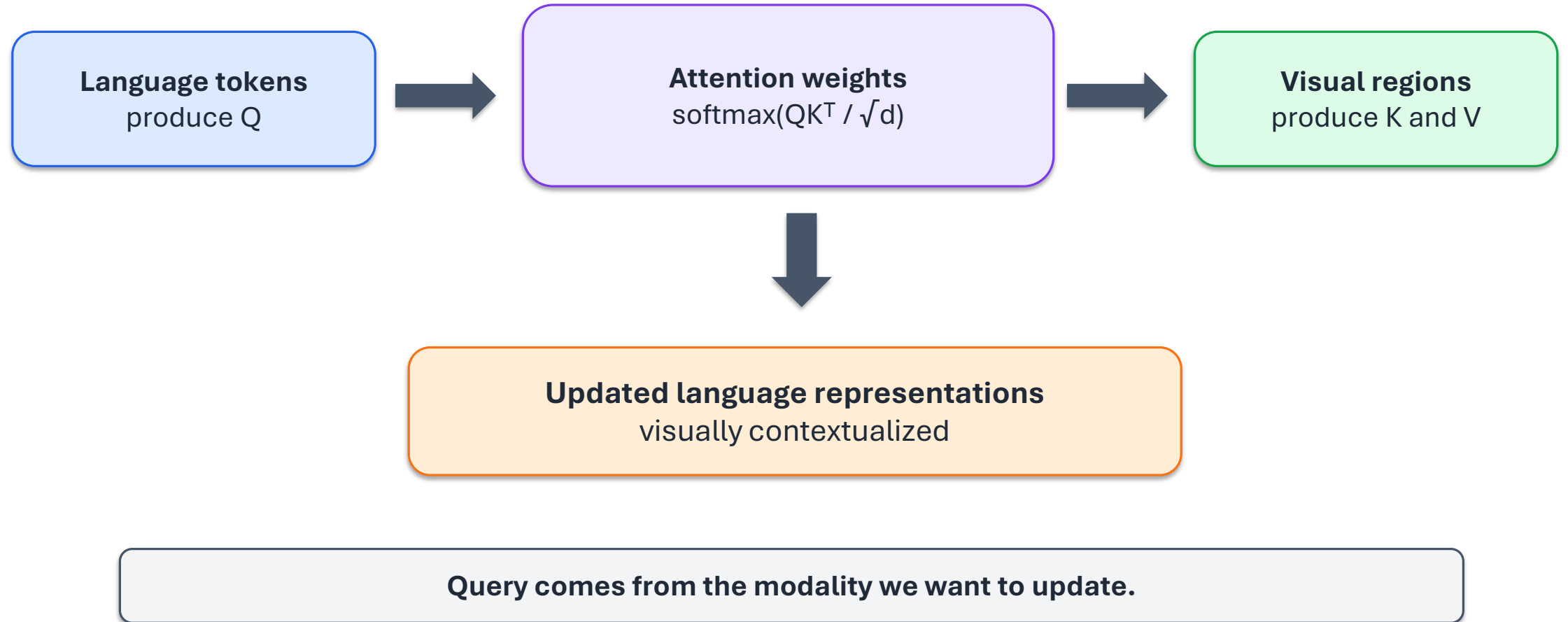


Cross-Modal Transformer Module ($\beta \rightarrow \alpha$)

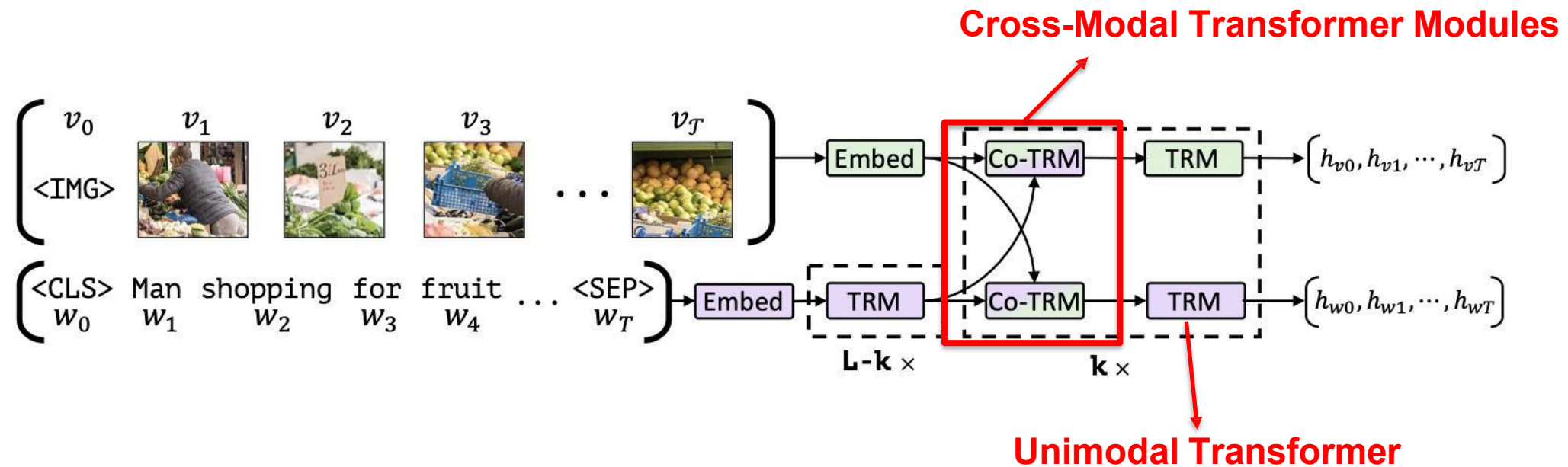


How Cross-Modal Attention Works

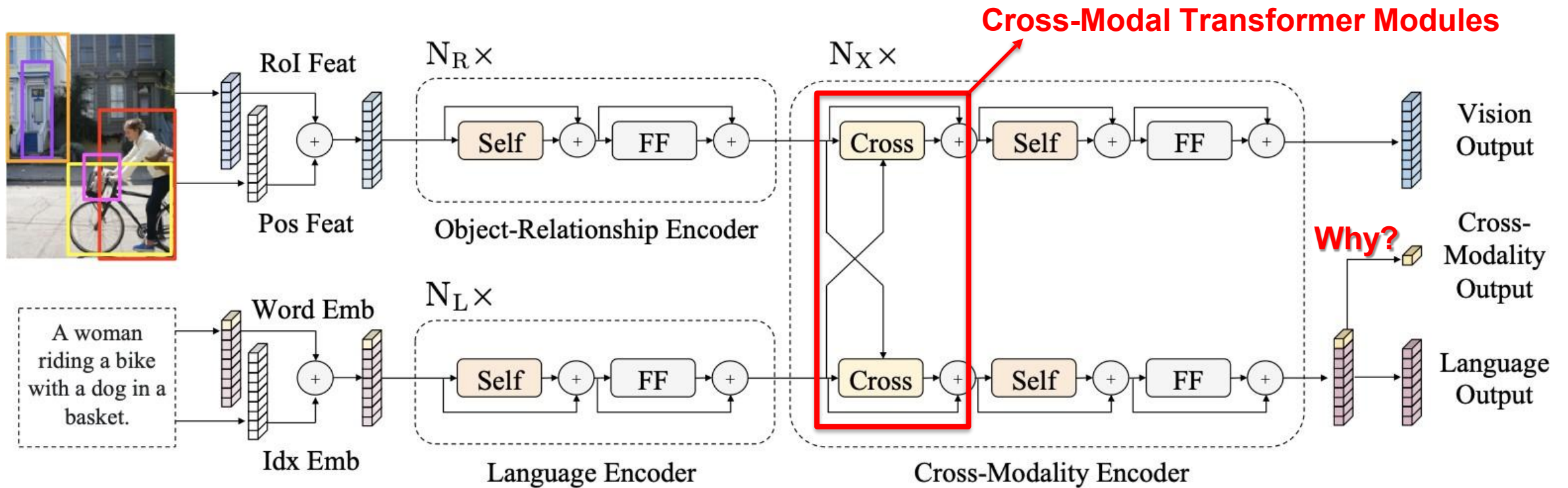
Example direction: Vision → Language



ViLBERT

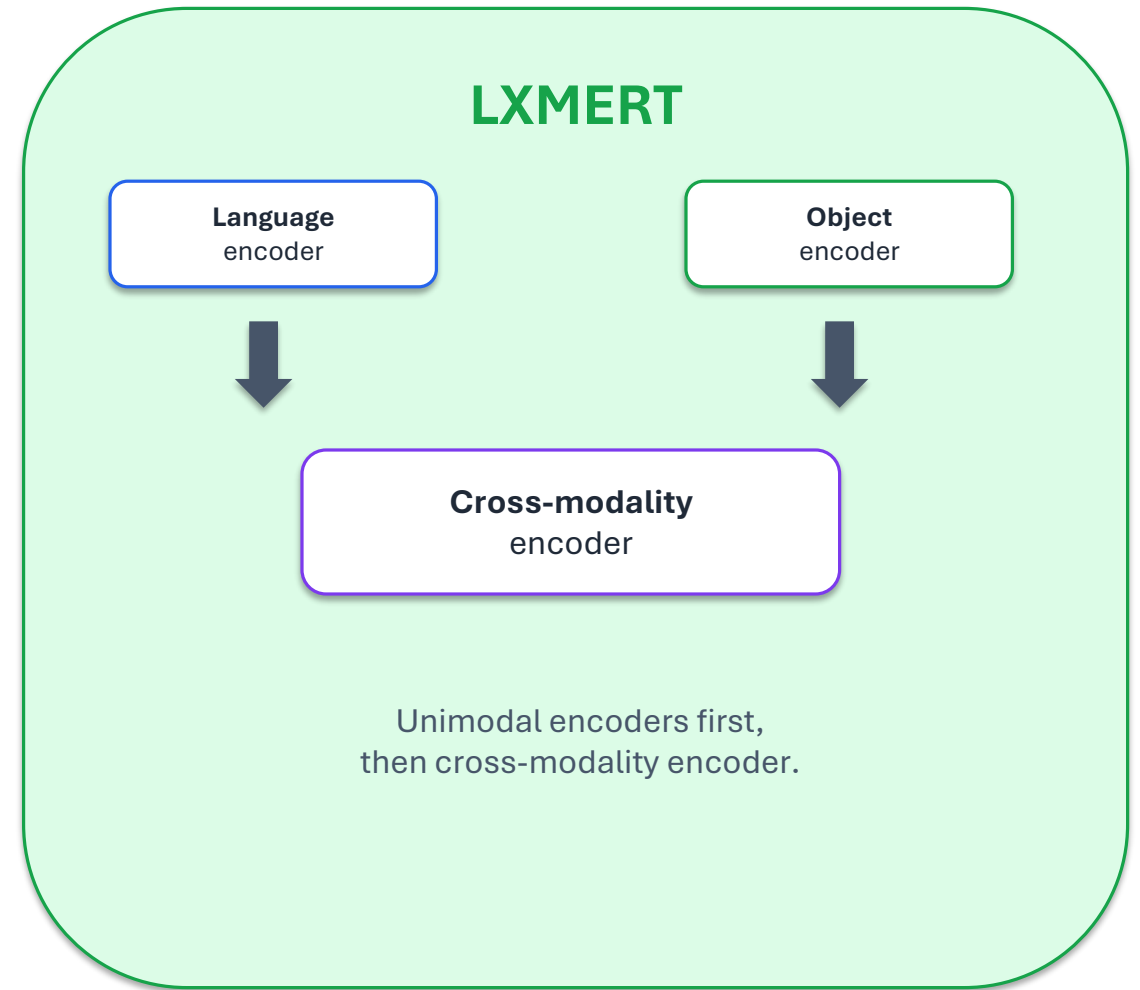
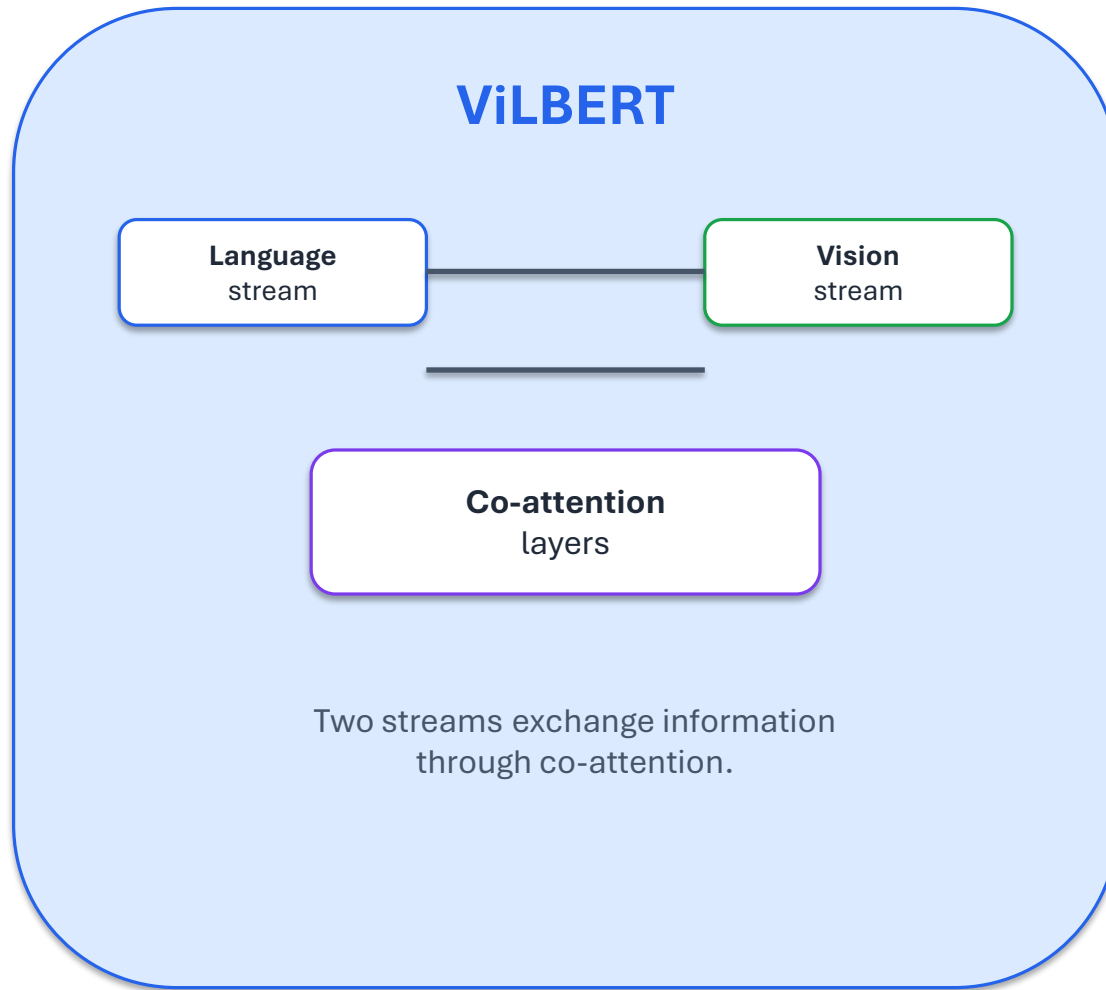


LXMERT



Tan, Hao, and Mohit Bansal. "Lxmert: Learning cross-modality encoder representations from transformers." *arXiv* (August 20, 2019).

ViLBERT vs LXMERT



Both: separate streams + cross-modal attention

Reminder: Modality-Shifting Fusion

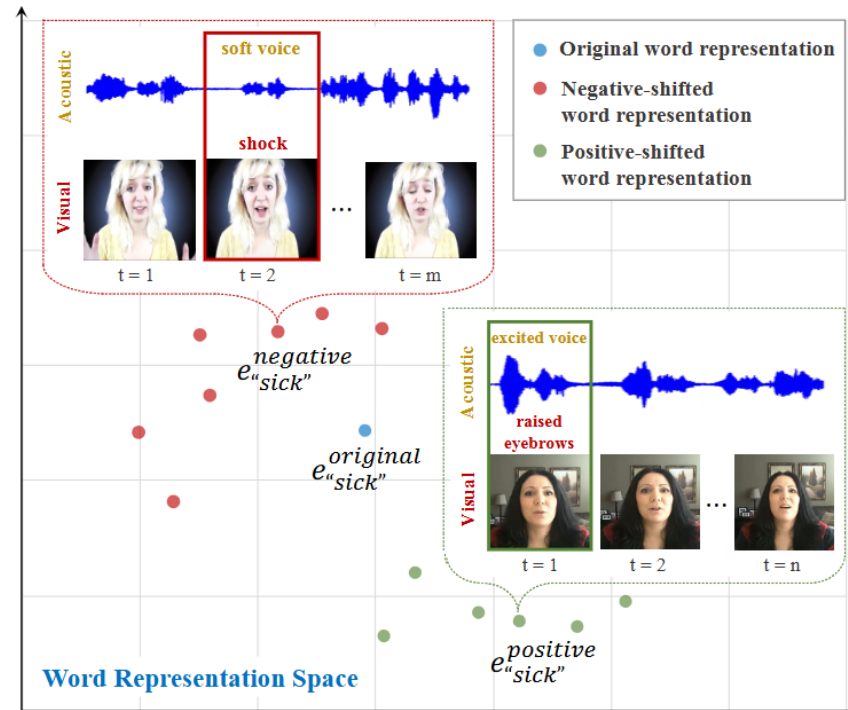
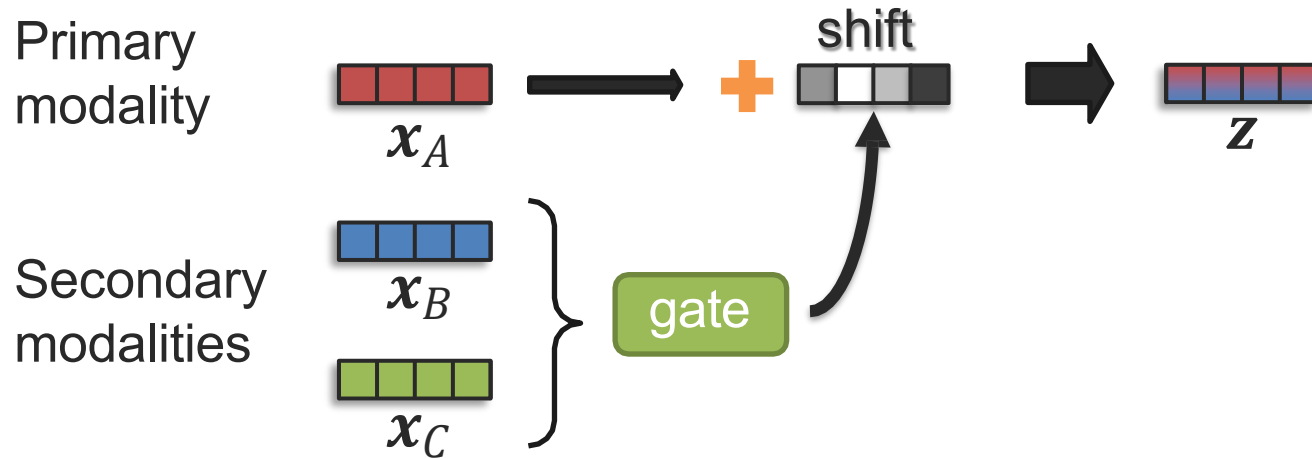


Figure 1: Conceptual figure demonstrating that the word representation of the same underlying word “sick” can vary conditioned on different co-occurring nonverbal behaviors. The nonverbal context during a word segment is depicted by the sequence of facial expressions and intonations. The speaker who has a relatively soft voice and frowning behaviors at the second time step displays negative sentiment.

Reminder: Modality-Shifting Fusion



Example with language modality:

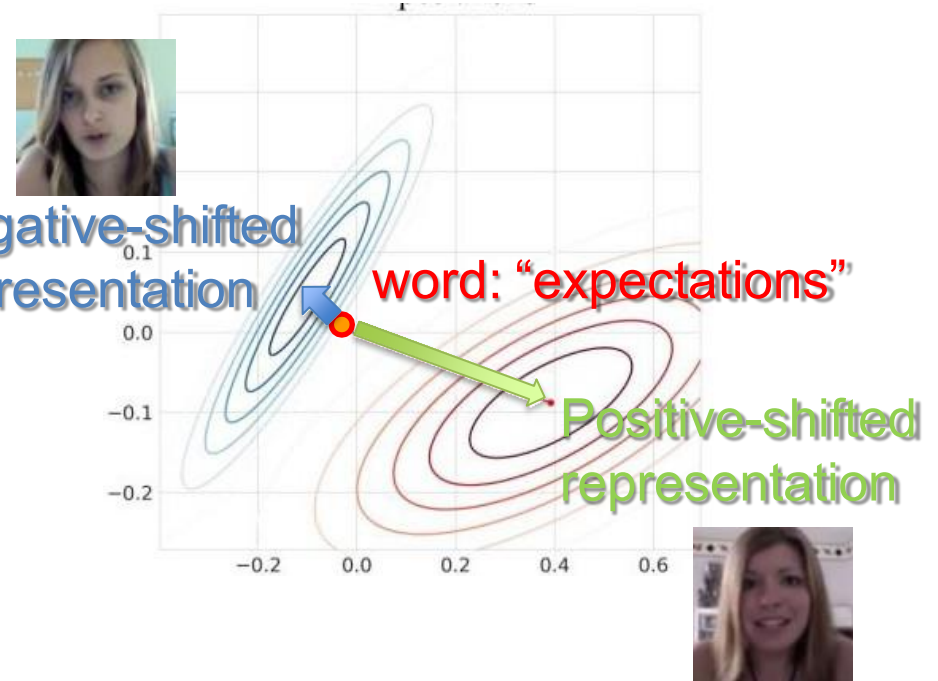
Primary modality: language

Secondary modalities: acoustic and visual

Negative-shifted representation

word: "expectations"

Positive-shifted representation



Reminder: Modality-Shifting Fusion

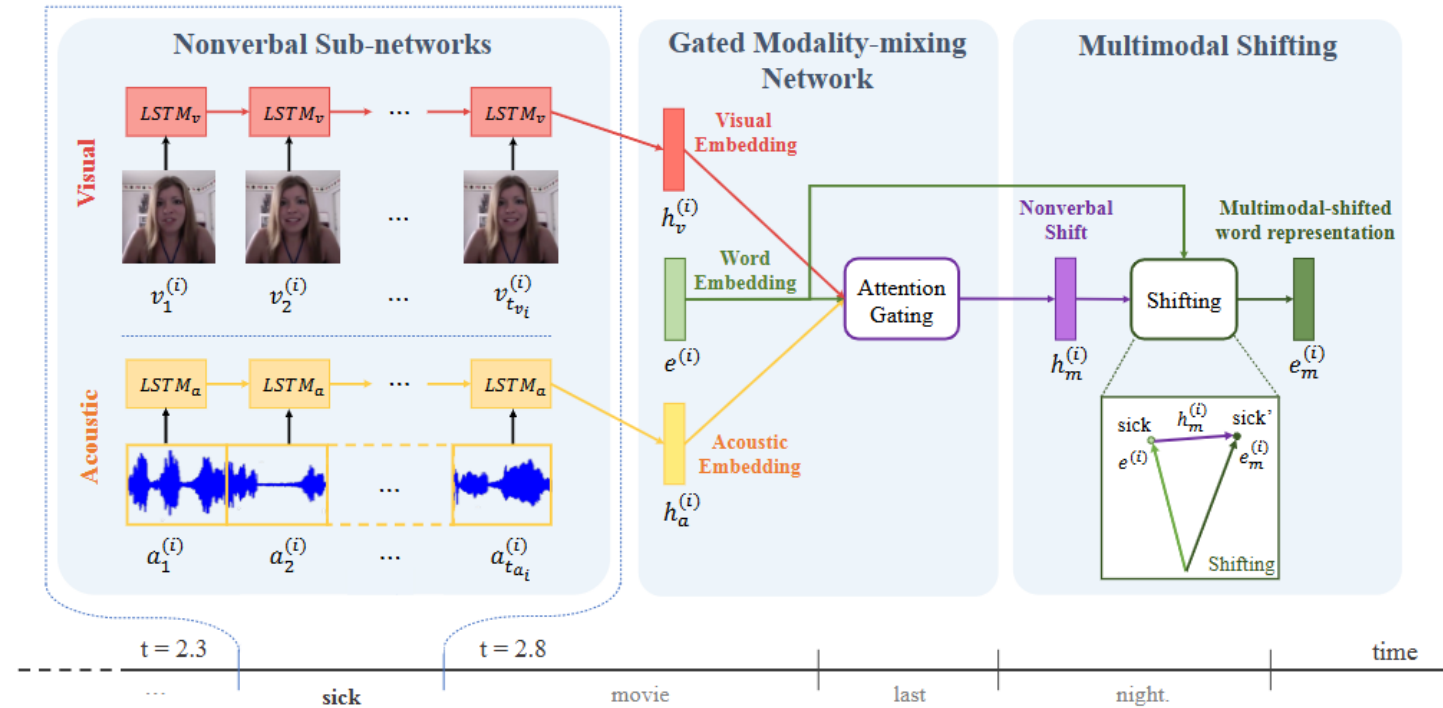
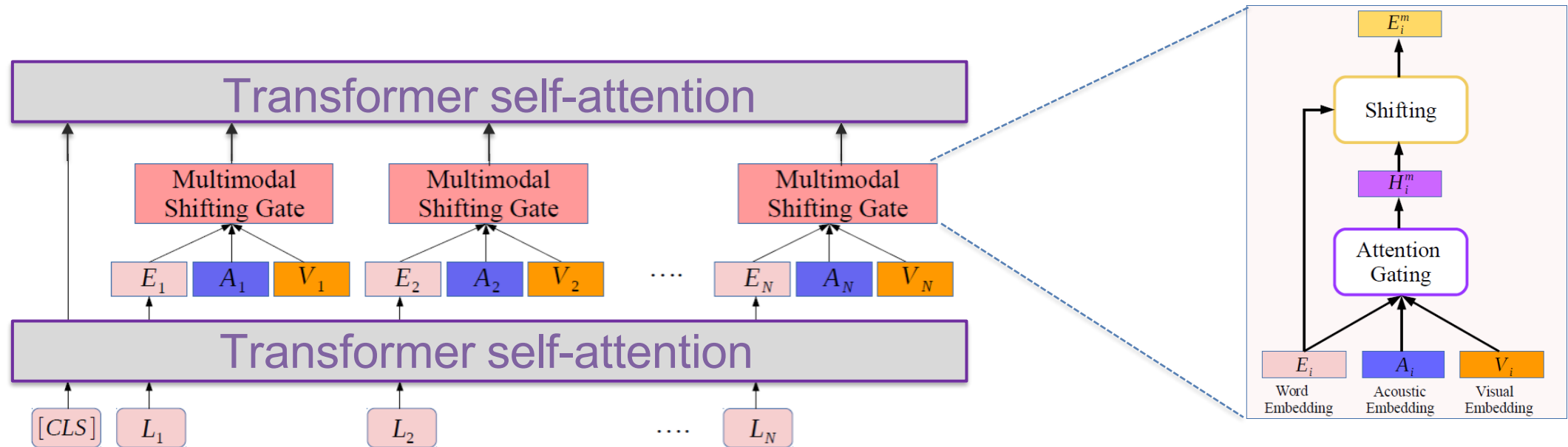


Figure 2: An illustrative example for Recurrent Attended Variation Embedding Network (RAVEN) model: The RAVEN model has three components: (1) Nonverbal Sub-networks, (2) Gated Modality-mixing Network, and (3) Multimodal Shifting. For the given word “sick” in the utterance, the Nonverbal Sub-networks first computes the visual and acoustic embedding through modeling the sequence of visual and acoustic features lying in a word-long segment with separate LSTM network. The Gated Modality-mixing Network module then infers the nonverbal shift vector as the weighted average over the visual and acoustic embedding based on the original word embedding. The Multimodal Shifting finally generates the multimodal-shifted word representation by integrating the nonverbal shift vector to the original word embedding. The multimodal-shifted word representation can be then used in the high-level hierarchy to predict sentiments or emotions expressed in the sentence.

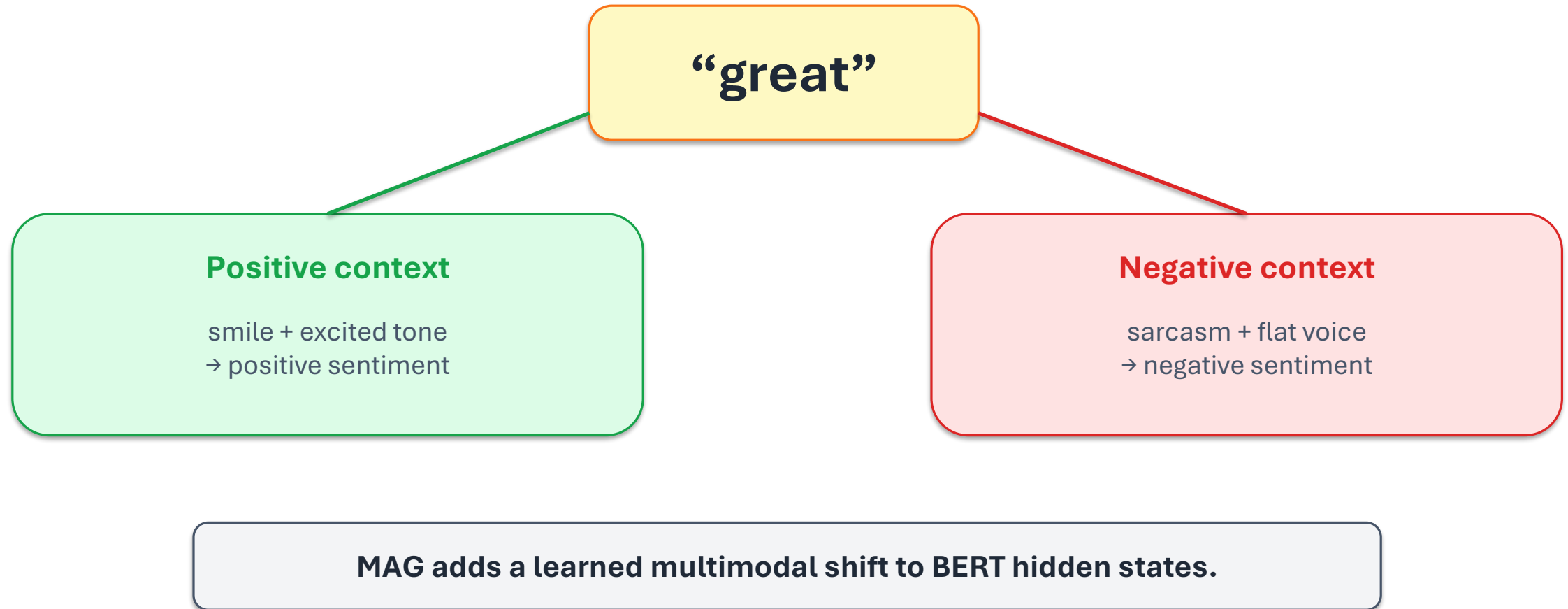
Modality-Shifting with Transformers

Multimodal Adaptation Gate (MAG) + BERT



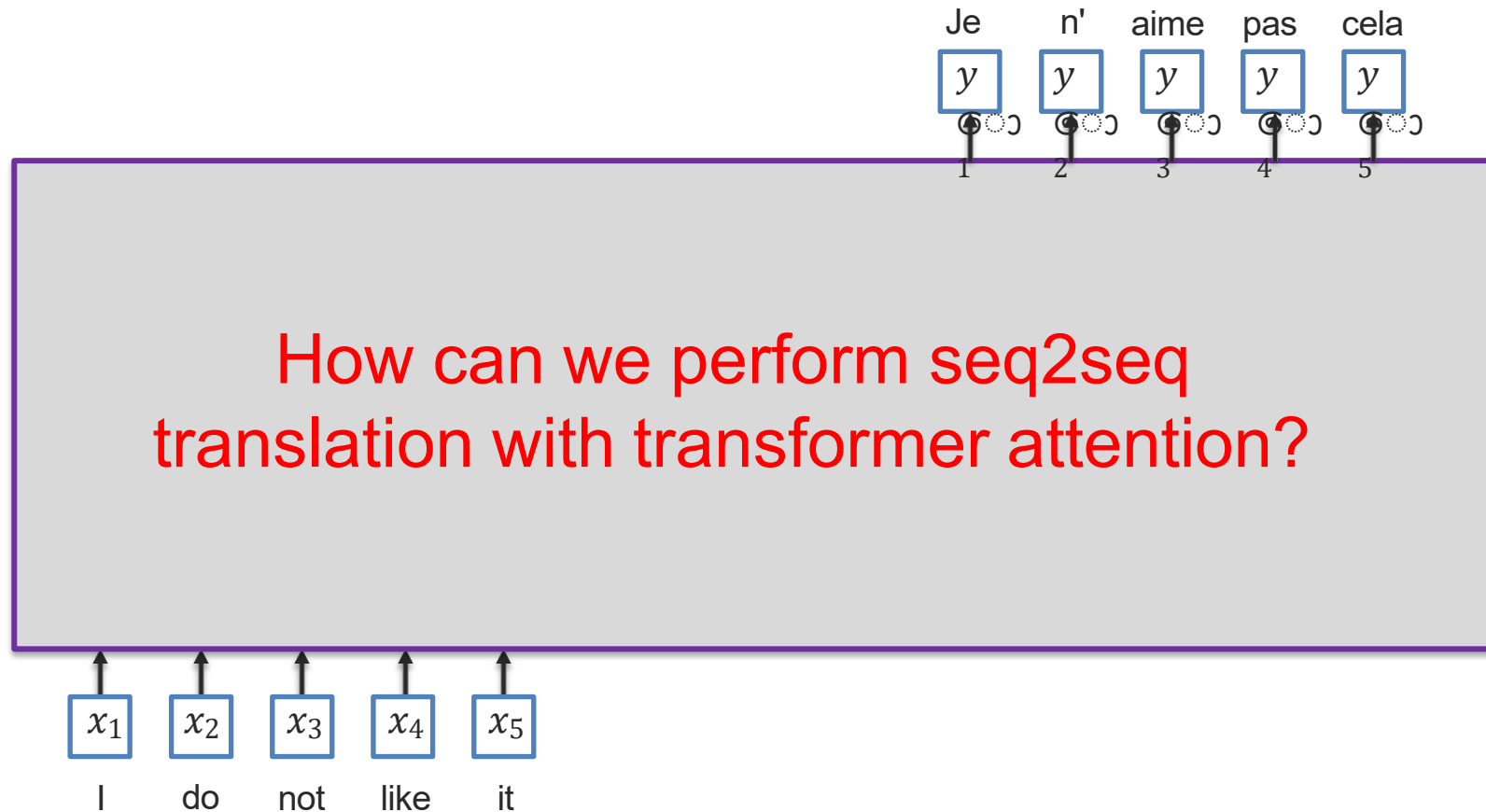
MAG-BERT: Modality Shift Intuition

Visual and acoustic signals can shift language meaning



Sequence-to-Sequence Using Transformer

Sequence-to-Sequence Modeling



Seq2Seq Generation Timeline

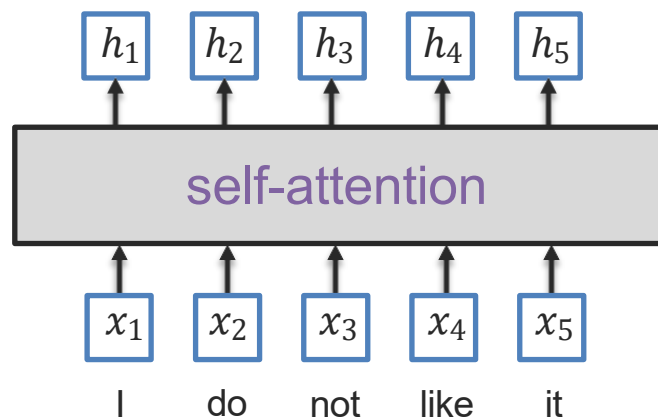
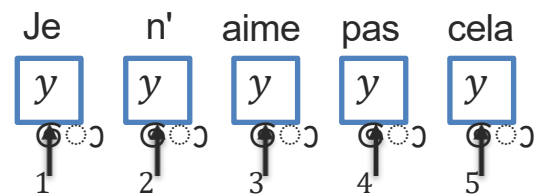
Source sentence: I do not like it

Target generation proceeds one token at a time

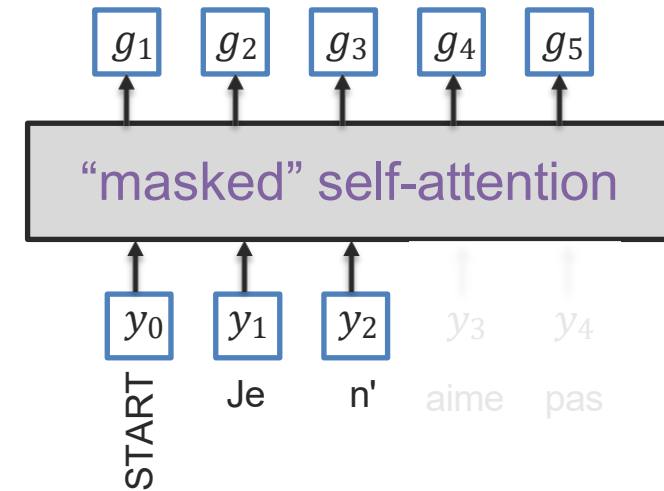
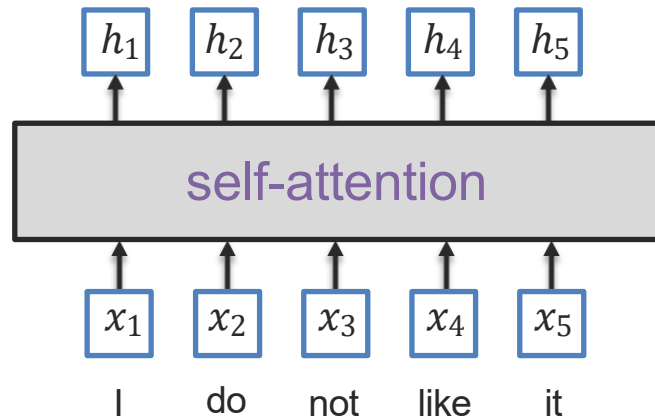
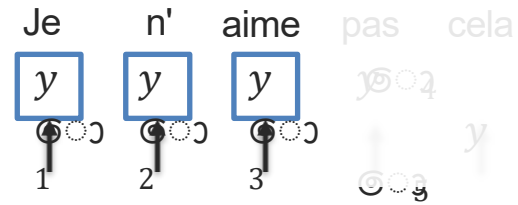


Each generated token becomes input for the next decoding step.

Seq2Seq with Transformer Attentions

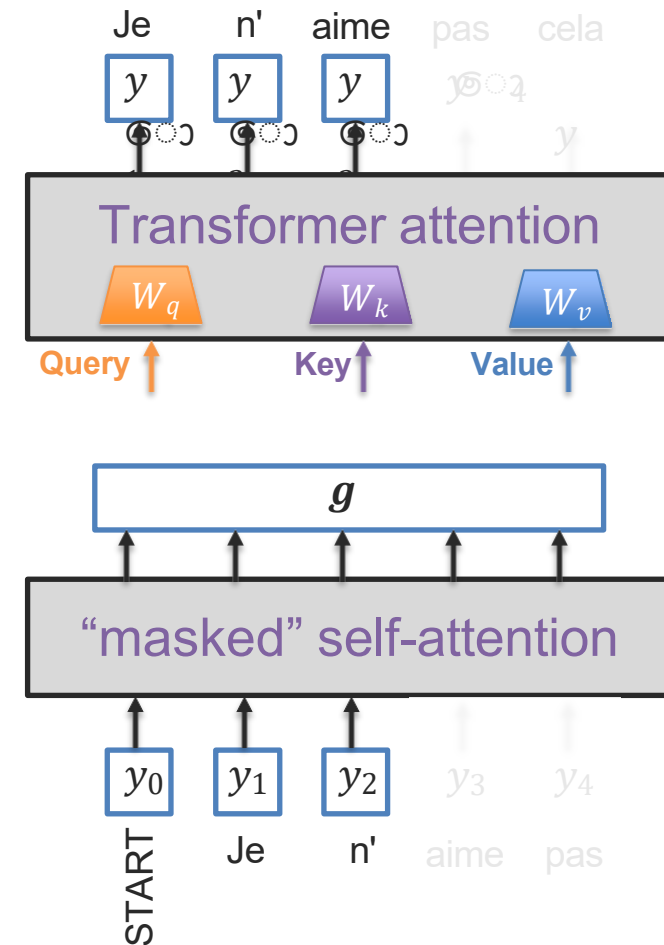
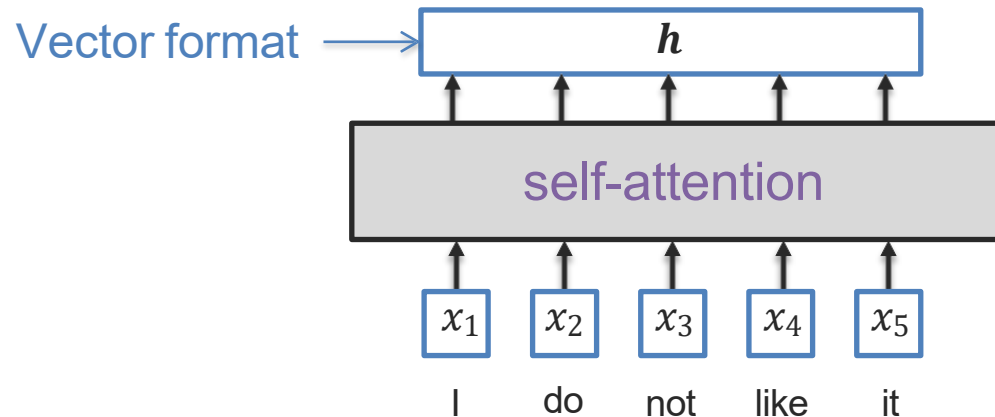


Seq2Seq with Transformer Attentions



Seq2Seq with Transformer Attentions

How should we connect the encoder and decoder self-attention to the transformer attention?



Masked Self-Attention vs Cross-Attention

Masked self-attention

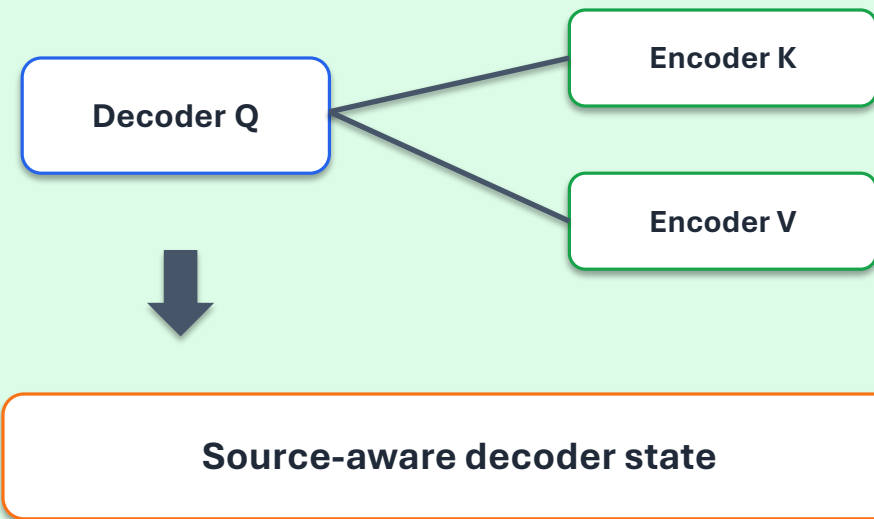
Decoder attends only to previous target tokens



future tokens blocked

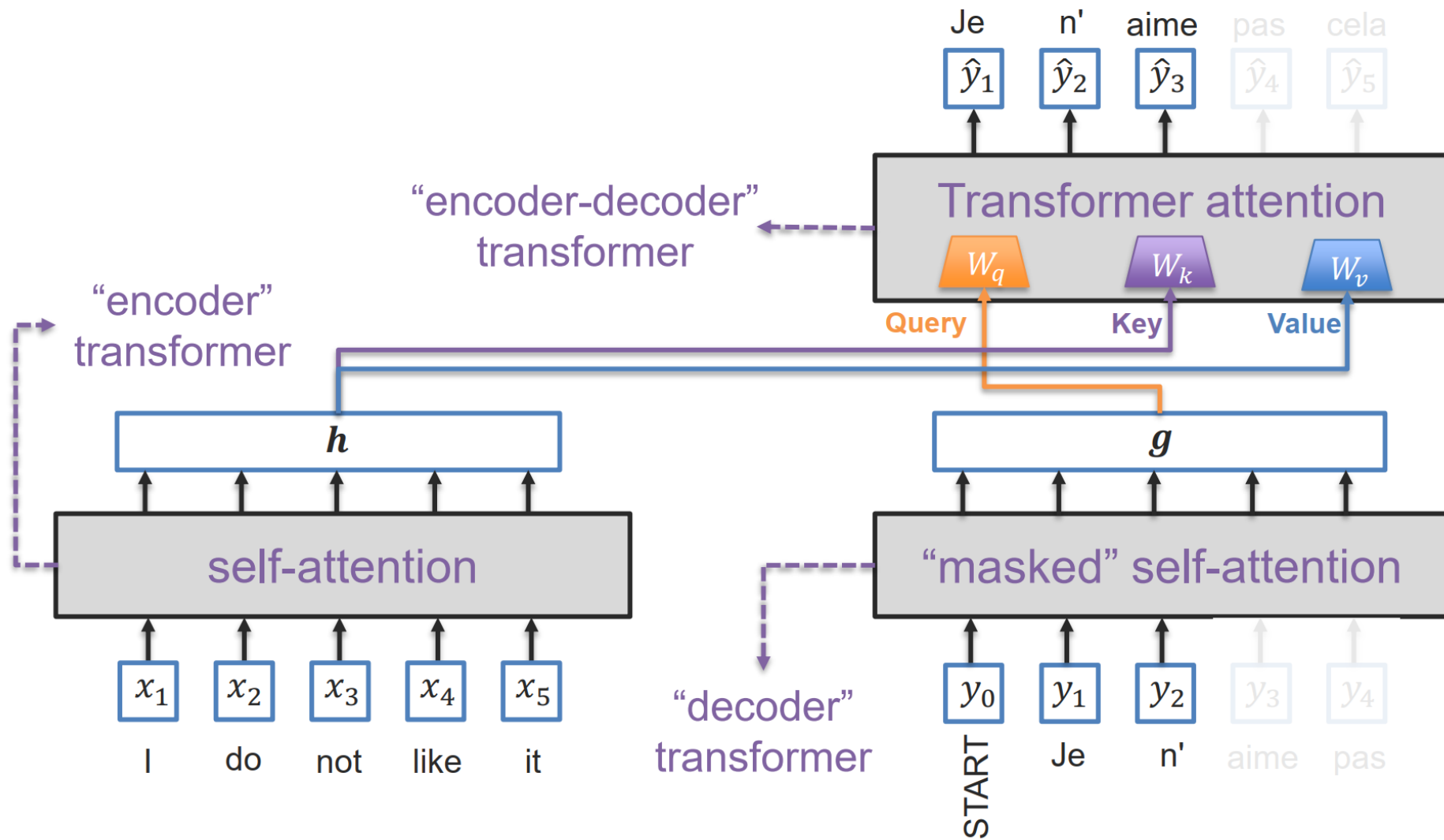
Encoder-decoder cross-attention

Decoder queries attend to all encoder outputs



Decoder = masked self-attention + cross-attention

Seq2Seq with Transformer Attentions



References

- Multi Modal Machine Learning, 11-777 • Fall 2023 • Carnegie Mellon University